

Delivering Power to the Space Elevator Climber

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For ISDC 2024

5/23-26/24

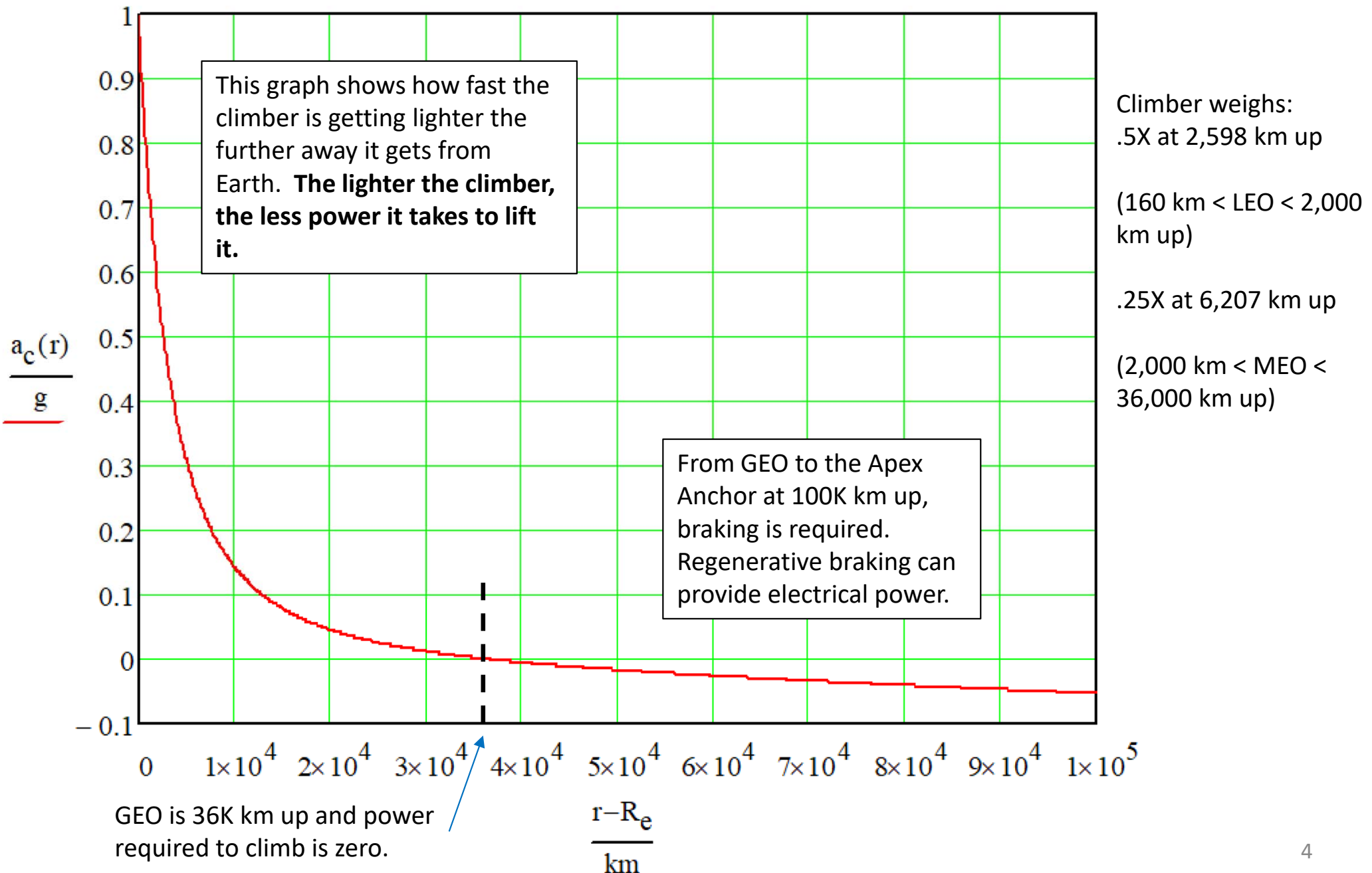
How to deliver megawatts to the climber?

- Because gravity is $1/r^2$, power need for the climber drops rapidly going to GEO
 - Power required to climb depends on the weight of the climber, which goes down with altitude
- A lot of power is needed to start from the surface of the Earth
- There are several zones the climber climbs through:
 - Acceleration
 - Constant velocity
 - Deceleration at GEO
- At launch an “extension cord” could be used to start the motors
- This talk will describe some features of a laser power beaming scheme from a few meters altitude to GEO

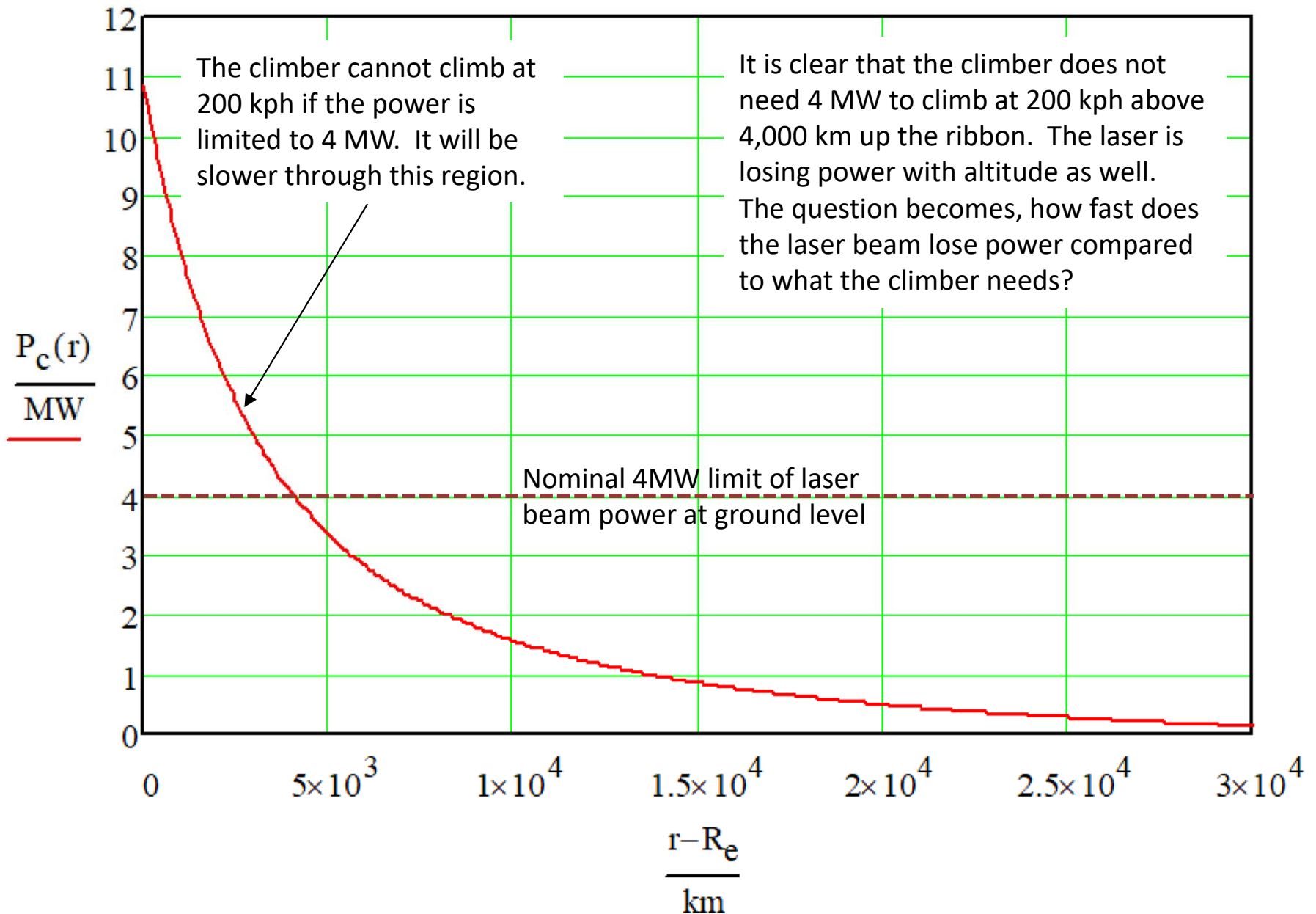
How much energy is required to put a kg in GEO?

- $\Delta E = E_{\text{GEO}} - E_{\text{Earth}}$ (Potential energy change)
- It takes 53.1 MJ/kg to boost objects up the tether to GEO
 - *This is the minimum energy required to put into every kilogram at GEO*
- Assume a 20 tonne climber, $m_c = 20,000$ kg
- A 20 tonne climber requires **1.062×10^{12} J** to get to GEO
- If the trip takes 8 days, (691,200 secs,) then the average power over the trip is 1.54 MW

Graph of acceleration felt by climber (ratio with Earth normal gravity) vs altitude (kilometers) from Earth's surface at the Equator



Graph of Mechanical Power (MegaWatts) vs Altitude up the ribbon (kilometers) for Constant Speed (200 kph) case for 20 tonne climber



What did Edwards propose in his book?

- In *The Space Elevator* by Edwards and Westling, they proposed to deliver 2.4 MW of laser power to a 20 ton climber
- A Compower FEL was selected to work with custom designed GaAs PV cells with 59% conversion efficiency, 82% filling factor and 540 kW/m² power density
- The PV array receiver on construction climbers ($900 \text{ kg} < m_{\text{climber}} < 20,000 \text{ kg}$) would start at 4 m diameter and go up to 10 m in diameter (proportional to climber mass)
- The transmitting laser would have a 12 m diameter mirror with adaptive optics to point the beam on the climber's receiving array

Conceptual design of a 20 tonne climber from the 2023 Study report showing different sizes of PV arrays

The light blue shell around the climber is a thermal radiator to dump waste heat to space.

This is the 20 tonne climber model shown in the ISEC study *The Climber-Tether Interface of the Space Elevator*, available at <https://www.isec.org/studies/#ClimberTetherInterface>

5 m diameter array

2.76 m diameter array

Motors driving each axle

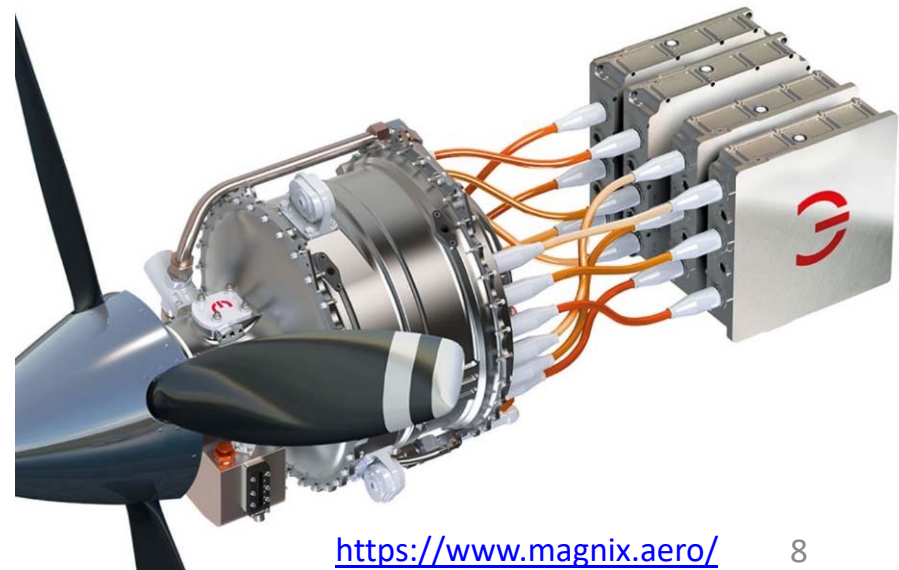
10 m diameter array

This climber is carrying two 5 tonne LOX tanks up to a future LOX depot at GEO.

I have never taken the time to develop the PV array support structure because the requirements are not known at this point. (Of course, this mass comes off payload mass.)

Climber Motor Characteristics

- The motors of the climber shown are extrapolated to be twice the torque and power of the Magni650 electric aircraft motor
 - Extrapolated motor torque = 5,628 N-m
 - Extrapolated motor power = 1.12 MW/motor
 - 10 motors/climber → 11.2 MW total electrical power at full torque and rated speed
 - Operating at 800 V, each motor is drawing 1.4 kilo-amps of current
 - 5 motors on each half-climber means that the bus on each half is carrying 7 kAmps of current



Three prong approach to design

1. Atmospheric transmittance/Diffraction limit

- The atmosphere is ~97 km thick, but the power beam has to reach GEO at 36,000 km
- 99.7% of the range is beam in vacuum
- This study determines the laser light wavelength that works best due to absorption and diffraction

2. Laser Technologies

- What laser technology?
- What wavelength?
- What maximum power?
- What beam characteristics?

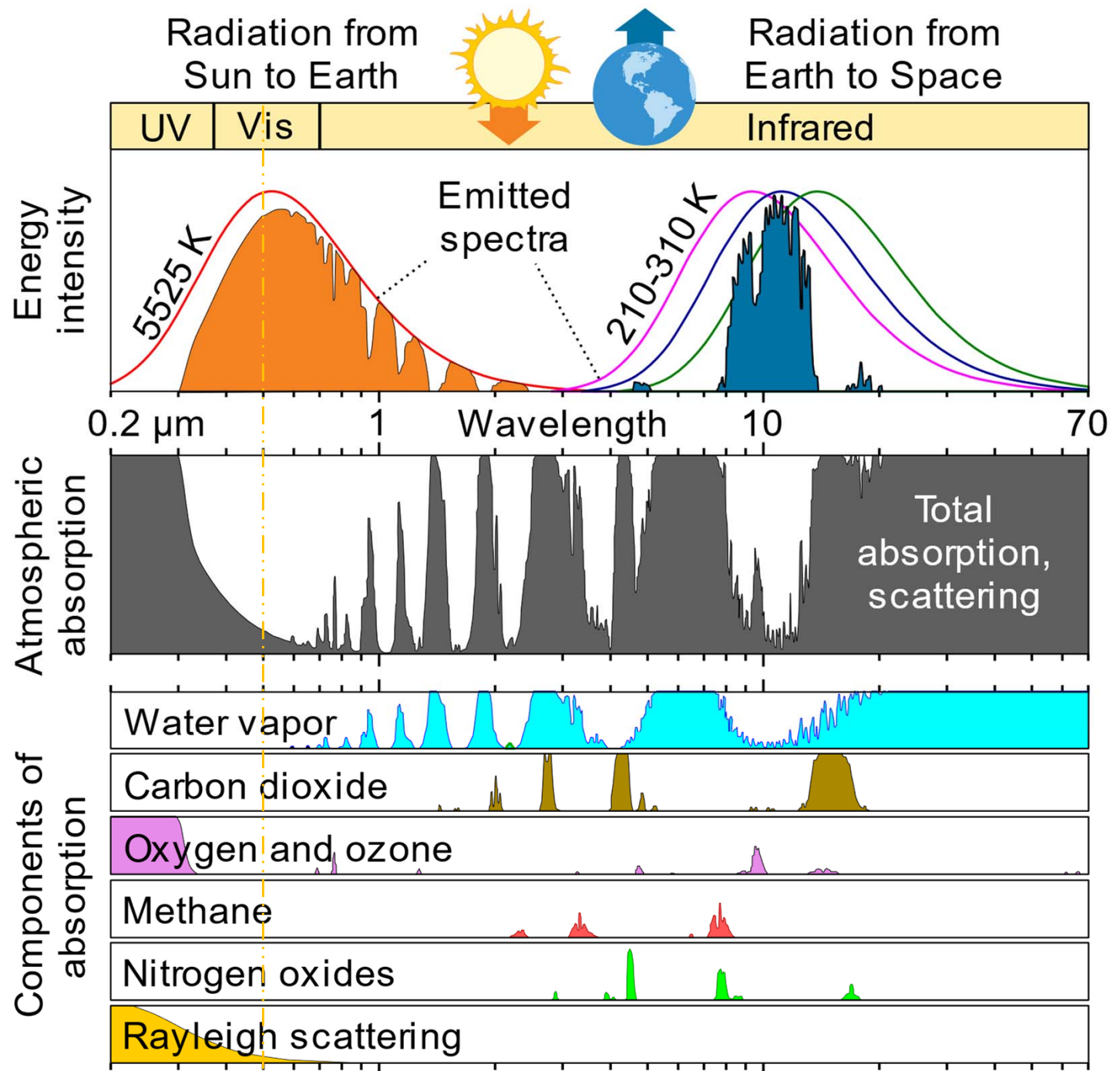
3. PV Receiver technologies

- Which PV technology gives the highest efficiency?

The Atmospheric Window (from the Wikipedia)

The orange phantom line is at .5 microns

There may be less scattering and absorption at a slightly longer wavelength, but then the beam diverges more at higher altitudes from diffraction. (See later slides.)



The diffraction of laser light propagating in vacuum

- The radius of a laser beam that starts out with a “waist” of w_0 gets larger as the beam propagates. The smaller the waist, the larger the beam spread with distance

- The equation is:

$$w(z) := w_0 \cdot \sqrt{1 + \left(\frac{\lambda \cdot z}{\pi \cdot w_0^2} \right)^2}$$

- w_0 = radius of laser transmitter
- $w(z)$ is the beam radius at altitude, m
- λ is the laser wavelength, .3-.84 μm
- z is the distance away from the laser, m

From the equation we see that the smaller the wavelength, the less divergence we get. Need to choose a frequency with a practical laser, matched to a PV array, that isn't absorbed by the atmosphere.

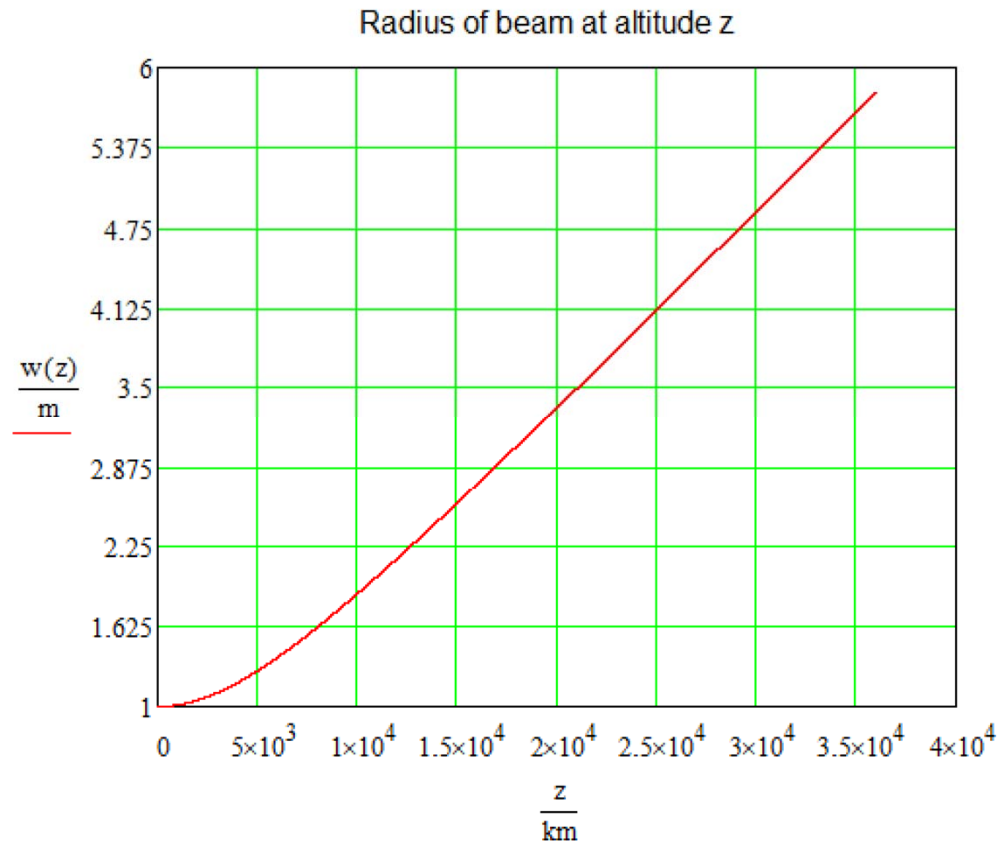
Start with a beam waist radius $w_0=1$ m (laser diameter of 2 m)

$\lambda = .5$ microns, green laser

The power intercepted by the climber's PV array as a function of altitude is the ratio of the PV area to the beam area:

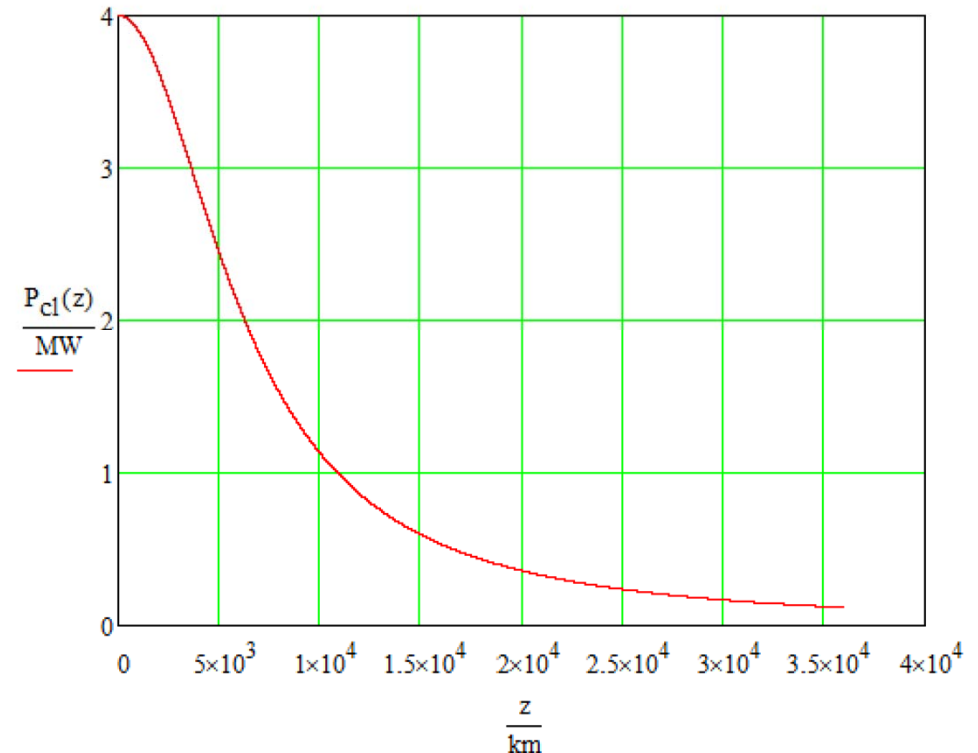
$$P_{cl}(z) := P_o \cdot \pi \cdot \frac{(R_{PV})^2}{A_{beam}(z)}$$

Power of laser beam intercepted at altitude z , assuming a 1 m radius PV array on the climber



A 2 m diameter laser beam at the surface of the Earth becomes ~12 m in diameter at the distance of GEO.

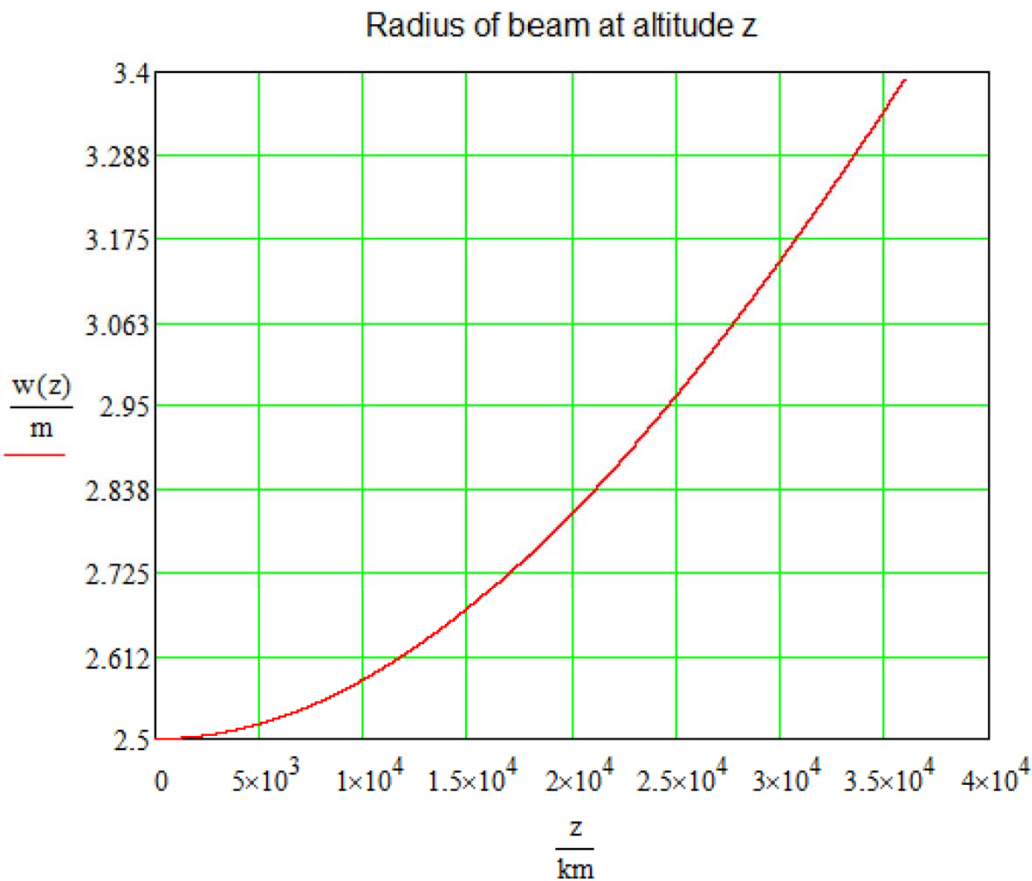
Since the PV array on the climber is a fixed diameter, less and less of the beam's power hits the array. Most of it misses at high altitude.



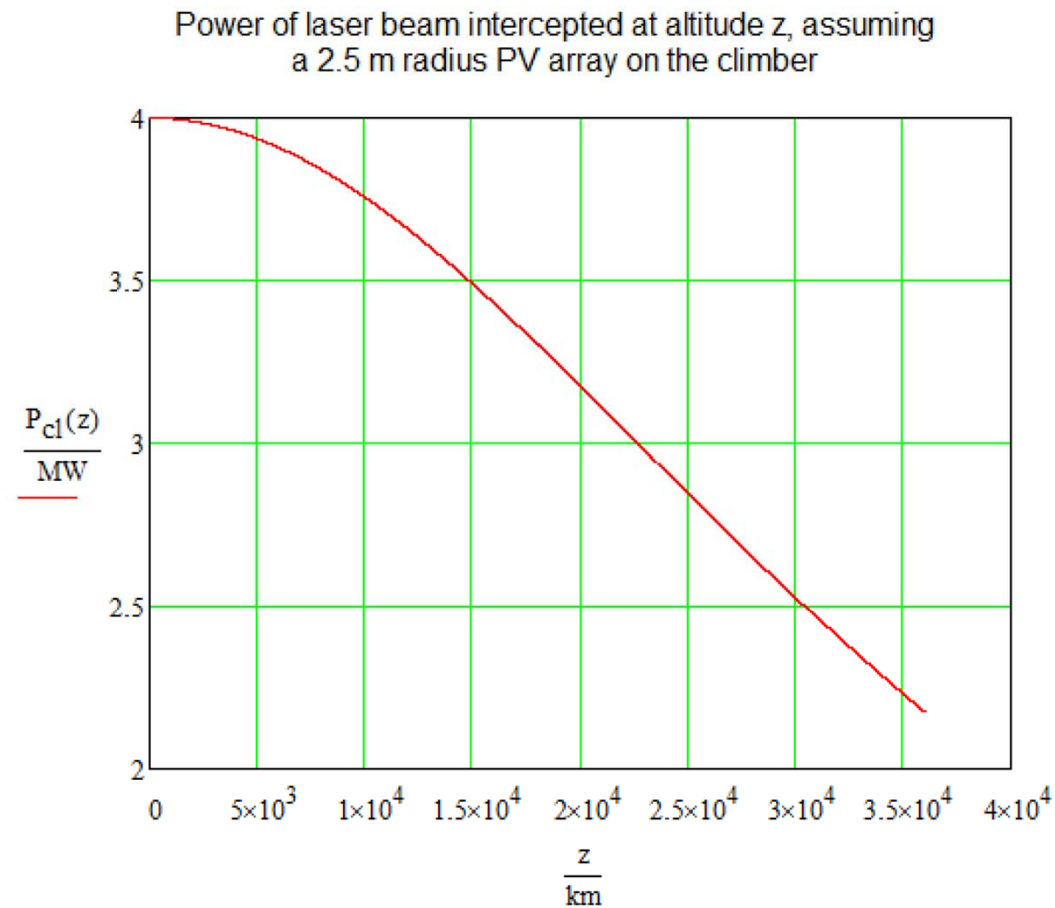
The power received by the climber drops 95% by the altitude of GEO because the beam has spread so wide.

Start with a beam waist radius $w_0=2.5$ m (laser diameter of 5 m)

$\lambda = .5$ microns, green laser



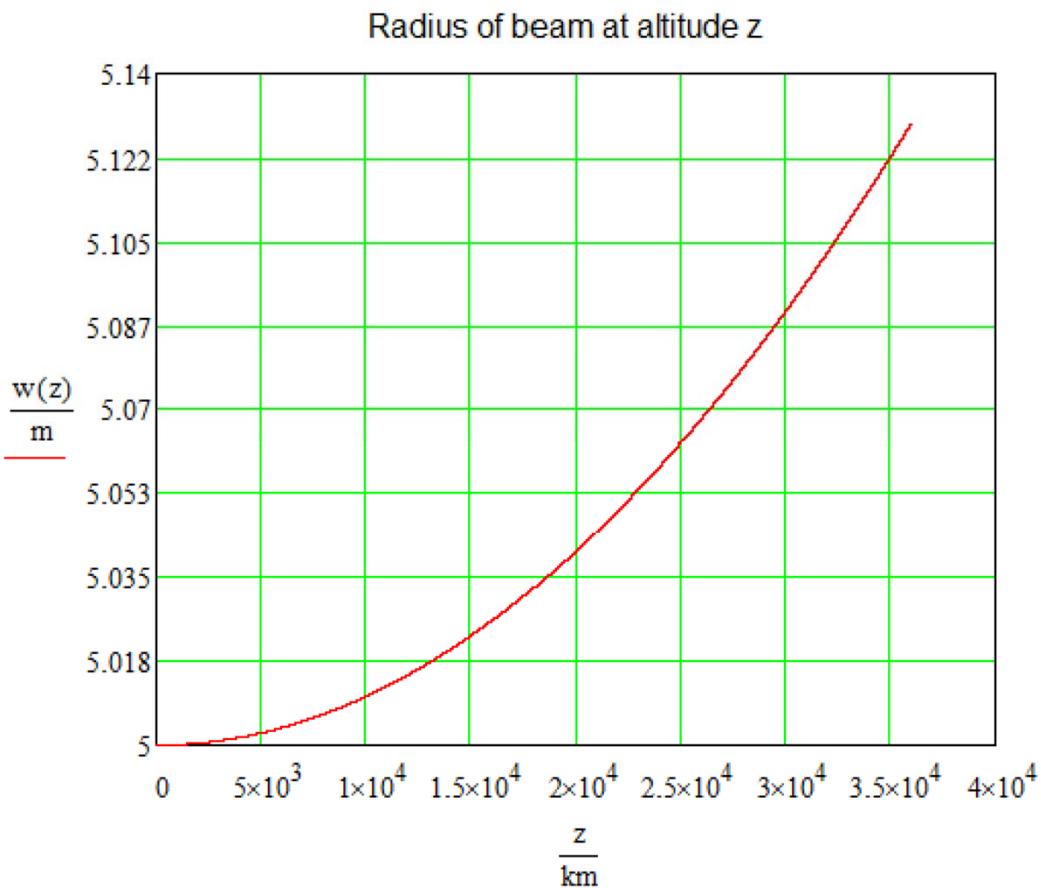
The beam spreads from 5 m in diameter to 6.8 m at GEO.



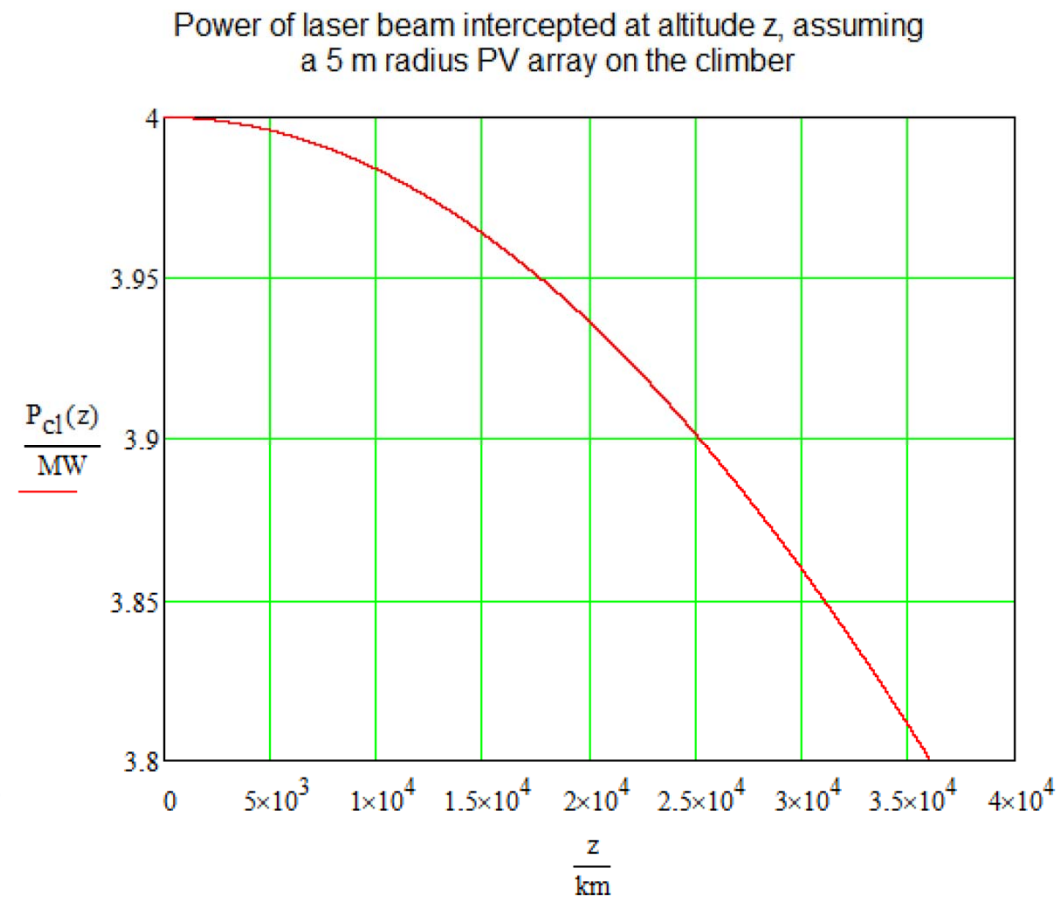
The power drops by 45% by GEO.
There is still over 2MW available of the 4 MW transmitted.

Start with a beam waist radius $w_0=5$ m (laser diameter of 10 m)

$\lambda = .5$ microns, green laser



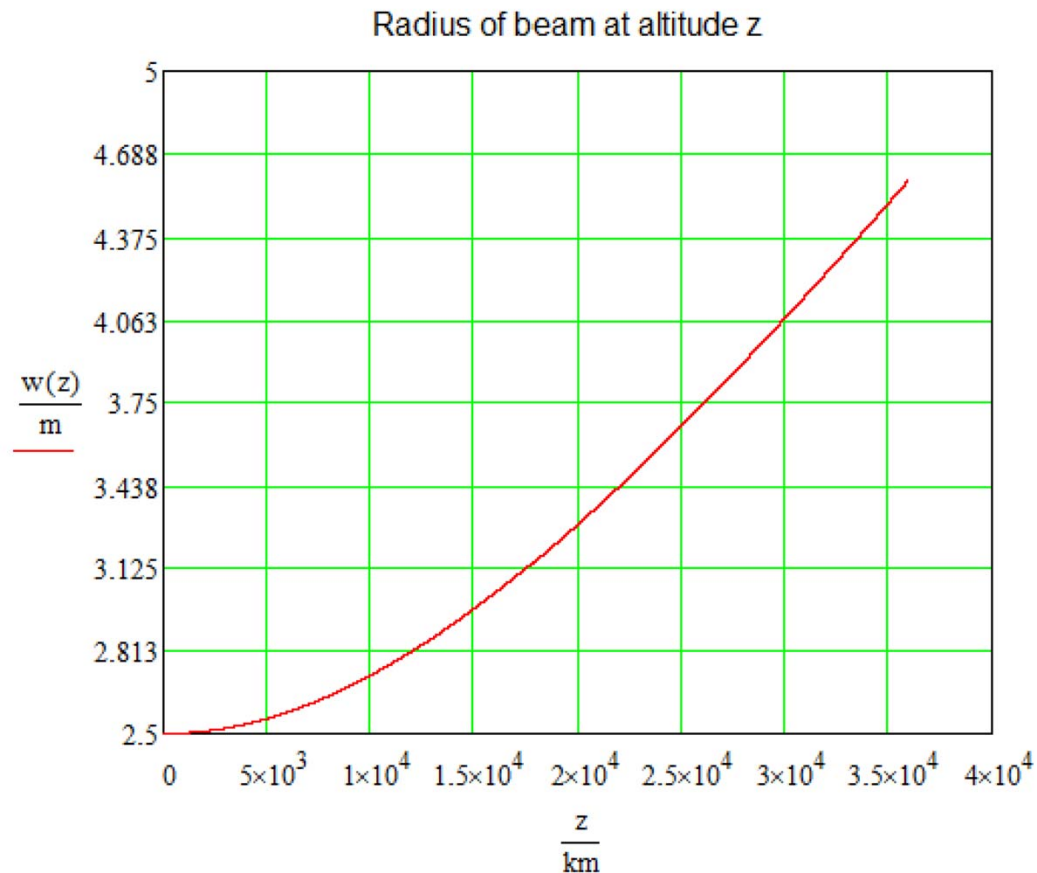
The beam spreads from 10 m in diameter to 10.3 m at GEO, very small beam spreading.



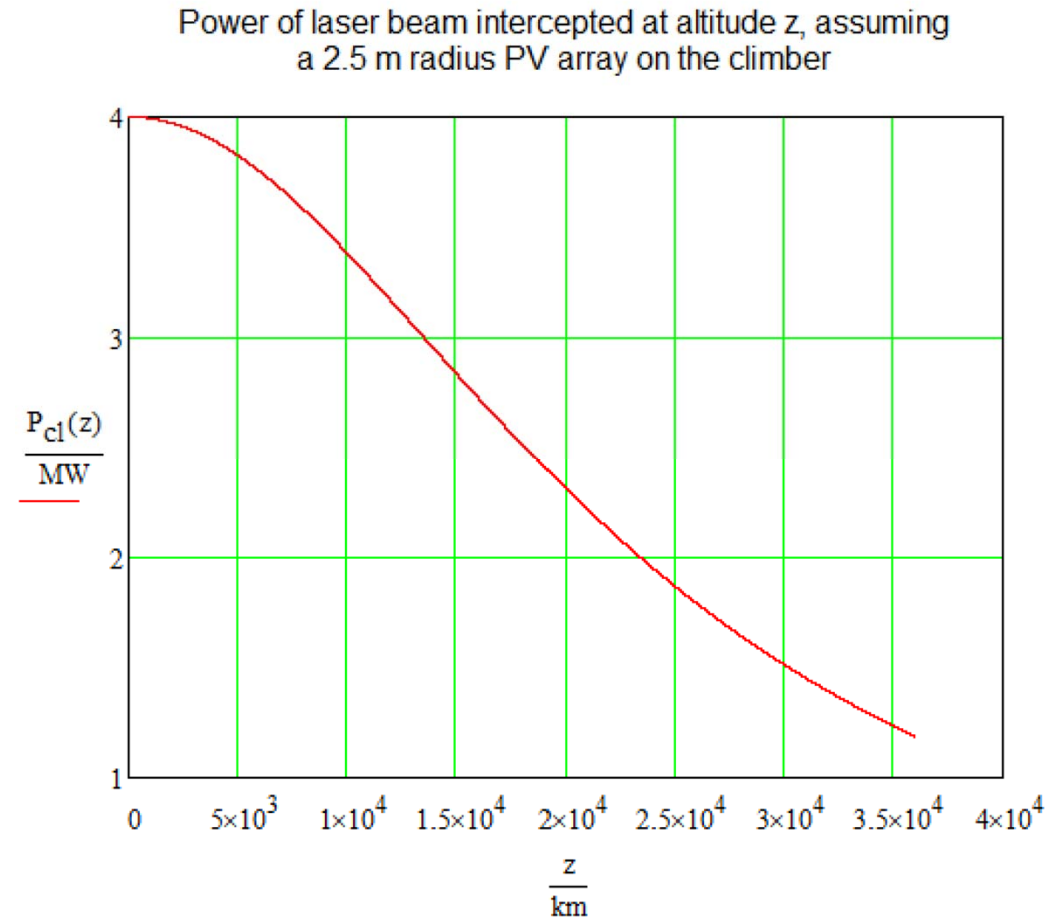
The power drops by 5% by GEO.
There is still over 3.8 MW available of the 4 MW transmitted.

Start with a beam waist radius $w_0=2.5$ m (laser diameter of 5 m), what is the effect of changing the frequency of the laser?

$\lambda = .84$ microns, middle of the range for GaAs PV cells



The beam spreads from 5 m in diameter to 9.1 m at GEO.

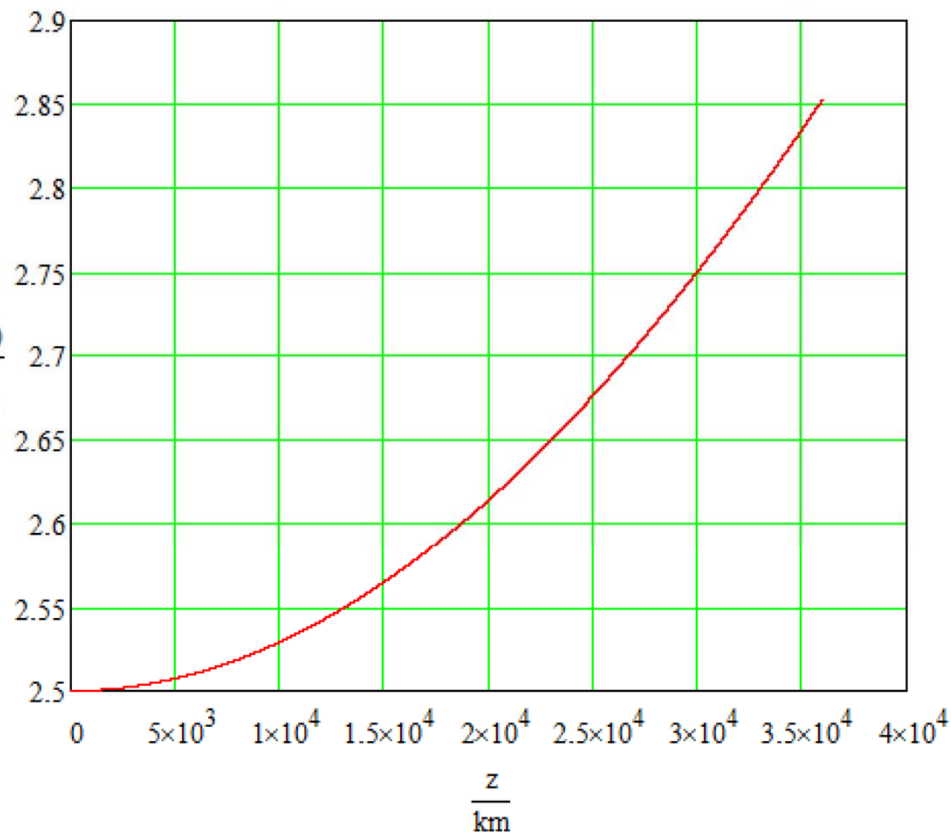


The power drops by 70.3% by GEO.
This is a large drop caused by making the wavelength larger than $.5 \mu\text{m}$.

Start with a beam waist radius $w_0=2.5$ m (laser diameter of 5 m),
what is the effect of changing the frequency of the laser?

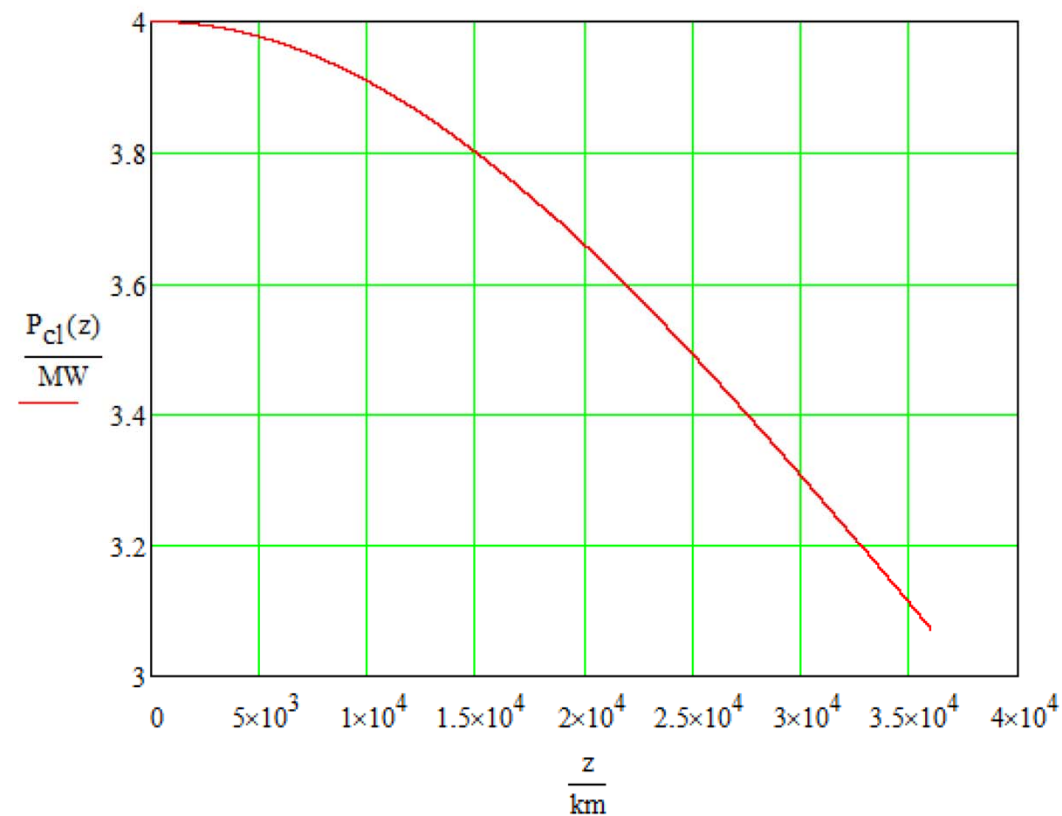
$\lambda = .3$ microns, UV laser with significant
atmospheric absorption

Radius of beam at altitude z



The beam spreads from 5 m in
diameter to 5.7 m at GEO.

Power of laser beam intercepted at altitude z, assuming
a 2.5 m radius PV array on the climber



The power drops by 22.5% by GEO.
This is a smaller drop than the 45%
drop for the $.5 \mu\text{m}$, but absorption
losses would be higher.

Conclusion about diffraction

- The larger the beam waist at the surface of the Earth, the more closely the divergent beam matches the PV array size at GEO
- Wavelength in the $.5 \mu\text{m}$ range diverges much less than infrared wavelengths
- A smaller diameter laser is cheaper and more feasible, so a laser array in the $\leq 5\text{m}$ diameter range may work fine
- The laser light that misses the climber could be collected on the bottom of the GEO node station for its use
 - The laser is delivering 110 kW/m^2 at GEO, compared to solar power at 1.37 kW/m^2

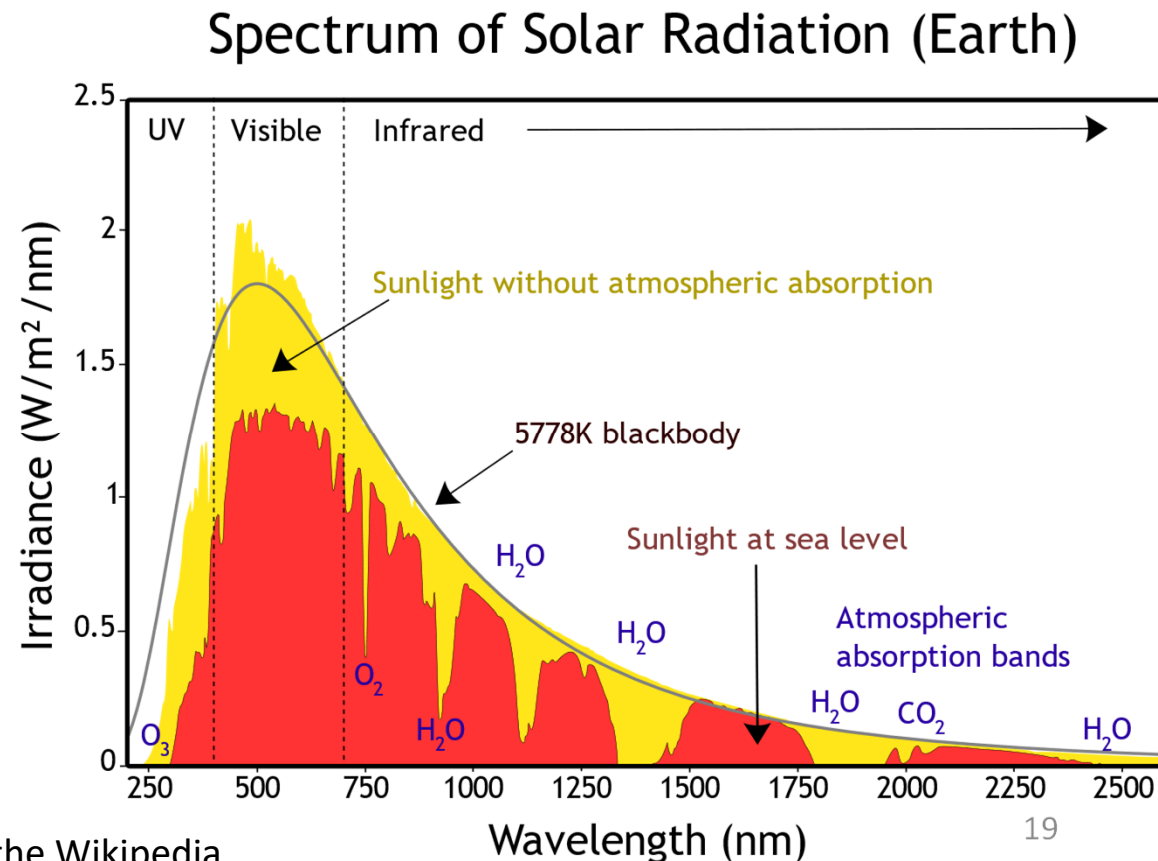
Atmospheric Transmittance

- Laser power at sea level is reduced in the atmosphere by **turbulence**, **absorption** and **scattering**
 - These are all complex processes to model
 - Weather is a significant factor
- How can we put a number on the maximum power loss caused by the atmosphere (.3% of the total travel of the power beam)?

Rough estimate of atmospheric total loss

- Solar irradiance outside the atmosphere at 500 nm wavelength: $1.9 \text{ W/m}^2/\text{nm} * 500 \text{ nm} = 950 \text{ W/m}^2$
- At sea level: $1.3 \text{ W/m}^2/\text{nm} * 500 \text{ nm} = 650 \text{ W/m}^2$
- $(950-650)/950 = .316$,
31.6% drop →
68.4% transmittance

- The atmosphere absorbs 31.6% of the laser power between ground and space
- This is for no clouds



Laser Technologies

- See the next slides for the range of possibilities
- A host of questions:
 - How to do Coherent Beam Combining to create a beam from a laser array so that the waist can be correlated with the array diameter? (This is a hard problem.)
 - How to incorporate Adaptive Optics?
 - How can the laser power be throttled as the need goes down without reducing beam quality?
 - Does it need to be tunable over a range of frequencies?

Possible green laser choices (Ref. 15)

- ***Argon ion lasers,***
 - based on amplification in an argon plasma (made with an electrical discharge)
 - The highest power can be achieved in green light at 514.5 nm.
 - It can exceed 20 W, but the power efficiency of such lasers is very poor
 - The beam quality can be close to diffraction-limited.
- ***Copper vapor lasers***
 - can emit relatively high powers at 510.6 nm.
 - They are based on a pulsed discharge in copper vapor and emit nanosecond pulses.
- ***Erbium-doped upconversion lasers***
 - based on erbium-doped fibers or bulk crystals can emit around 550 nm, typically with some tens of milliwatts of output power and with high beam quality.

More possible green laser choices

- *Dye lasers*

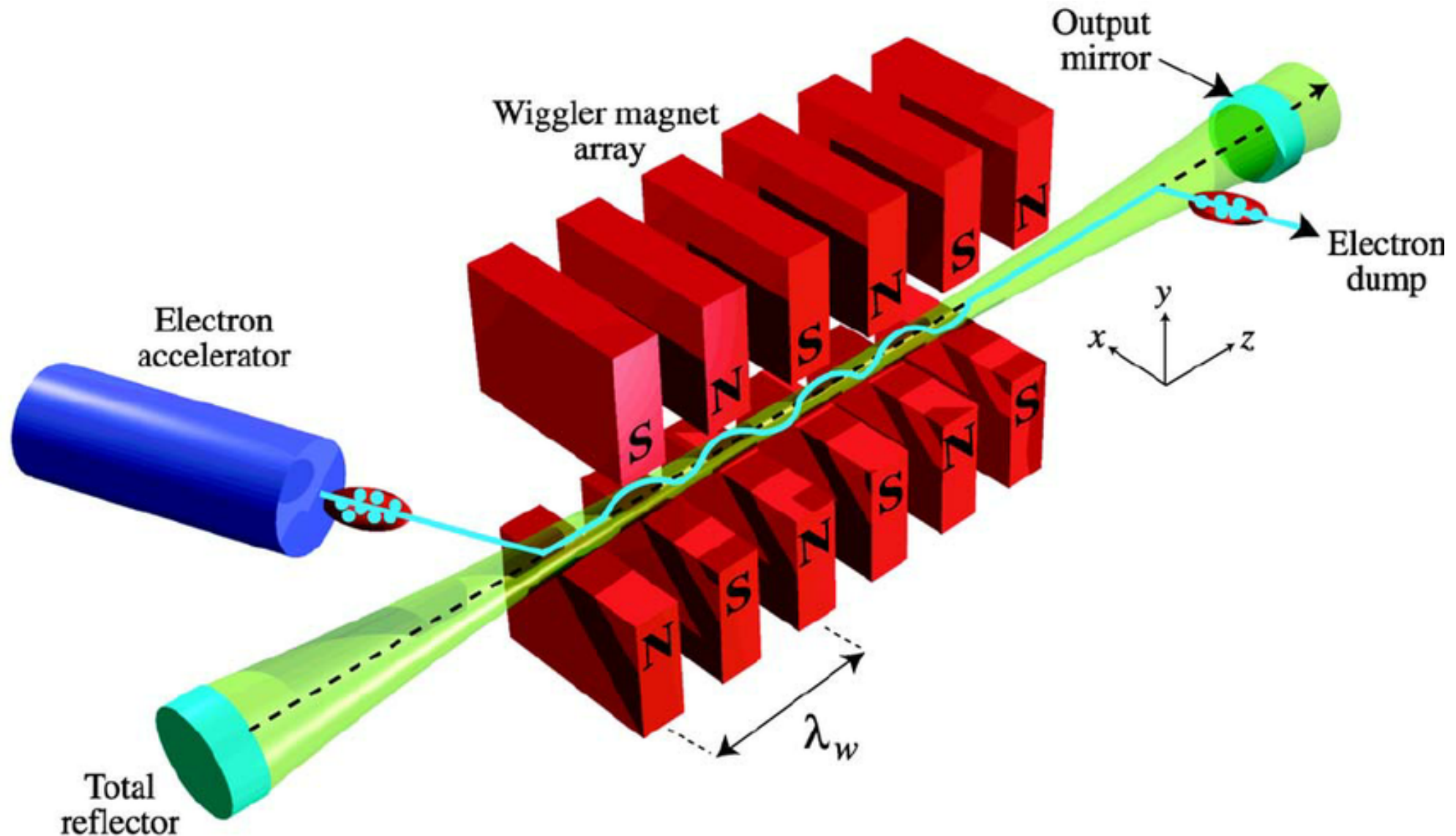
- can be operated in the green spectral region with wide tunability.



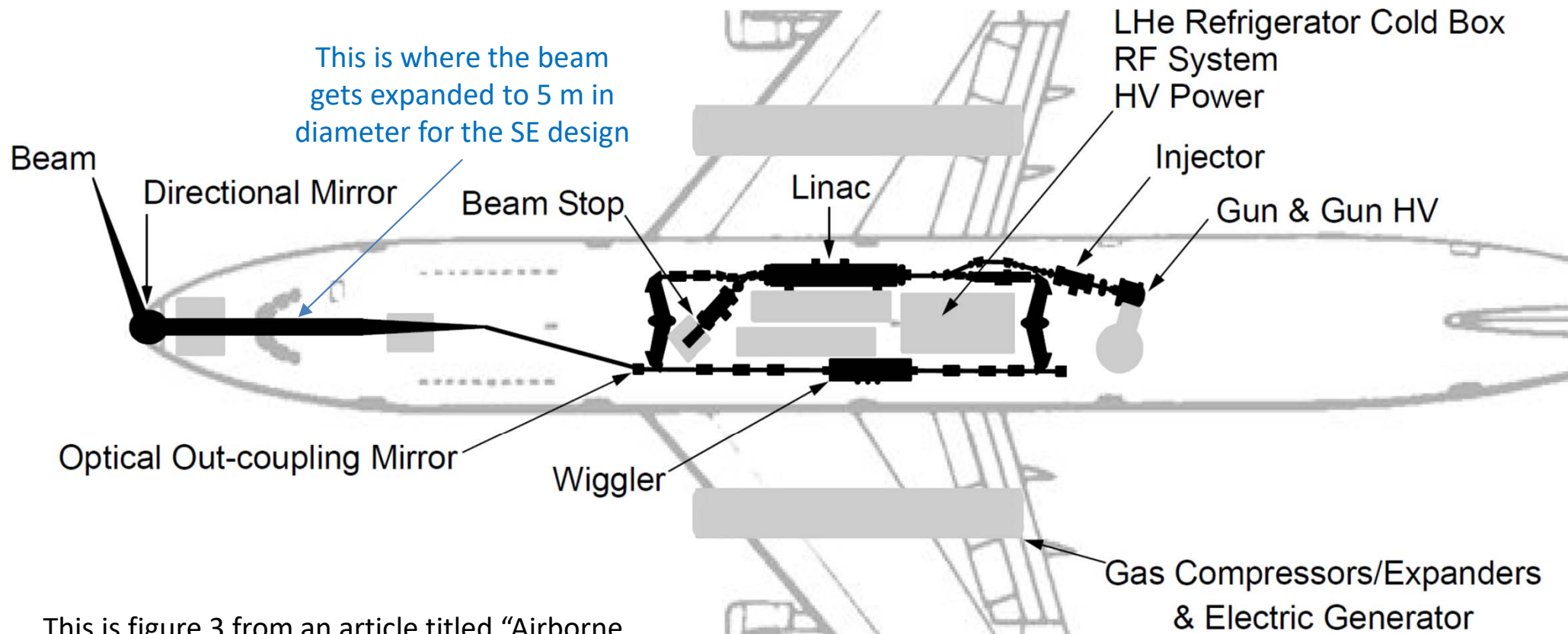
Figure 2: A high-power thin disk laser with intracavity frequency doubling generates 1.3 kW of green light with excellent efficiency. Source: Max Kovalenko, *Institut für Strahlwerkzeuge, Stuttgart*.

Green [laser diodes](#) (or other green [semiconductor lasers](#)) are difficult to produce – even more difficult than [blue](#) ones. They can emit only a few milliwatts and have relatively short lifetimes, compared with other laser diodes [\[5\]](#). However, there has been encouraging progress recently [\[10, 11\]](#). (From Ref. 15)

Free electron lasers can emit green light



Compact MW FEL for aircraft



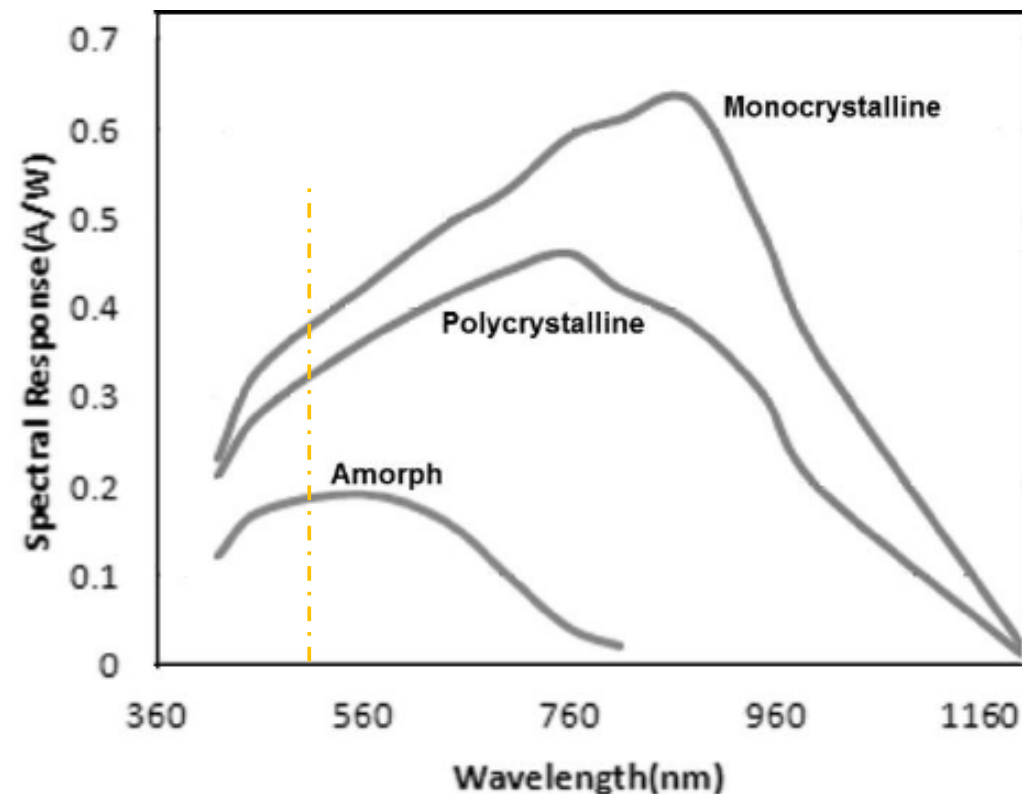
This is figure 3 from an article titled "Airborne megawatt class free-electron laser for defense and security" (Reference 17)

It was designed 20 years ago. Plasma wakefield technology may make the electron accelerator much smaller by improving energy transfer to the electrons by 400-500 times better than general linac designs.

This FEL was designed to be mounted on a 747 and use the jet's engines for power.

PV Receiver Technologies, 1

- Polycrystalline silicon
 - >20% efficiency conversion, Reference 3
- Single crystal silicon
 - >25% efficiency conversion, Reference 3



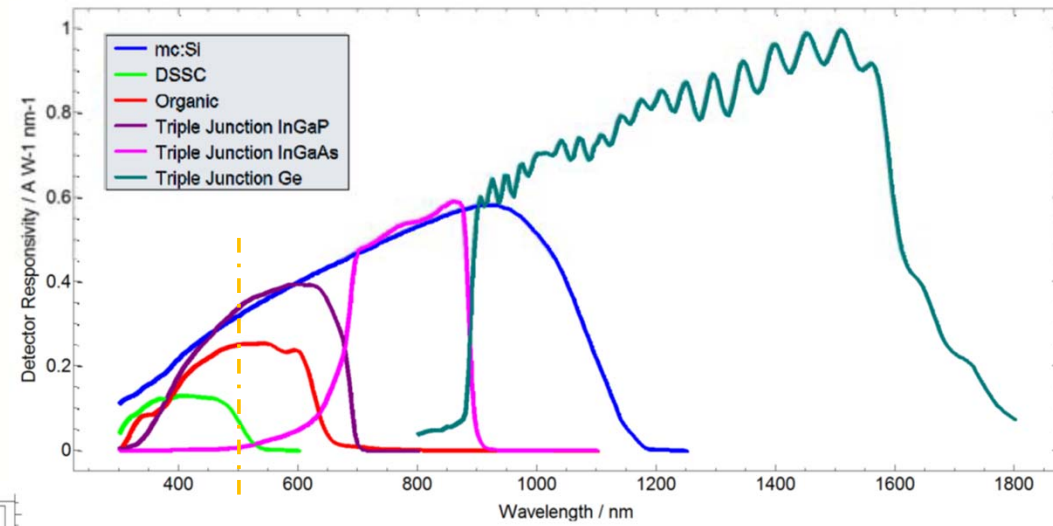
The spectral response of either polycrystalline or single crystal is peaked at 760 nm or higher, wavelengths that are too long and cause greater divergence of the beam.

(The orange line in these graphs shows the 500 nm wavelength.)

PV Receiver Technologies, 2

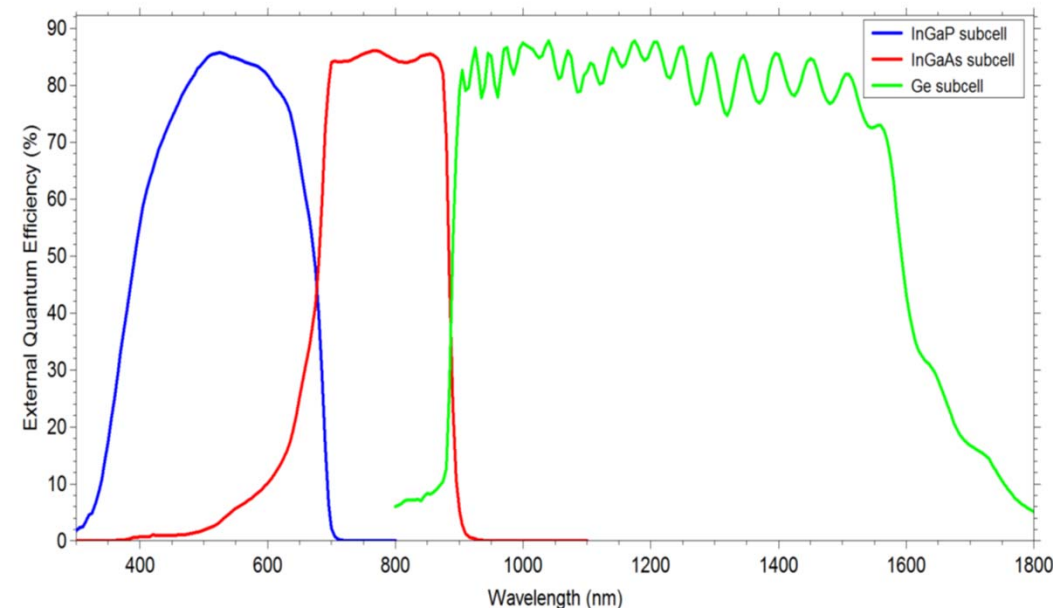
- Multijunction Solar cells (CPV)
 - 46% Efficiency in the lab, 30-40% commercial, Reference 5
 - Need cooling
 - Highest power-to-weight ratio (commonly used in aerospace applications)

External quantum efficiency and Spectral response of a number of multijunction cells from Ref. 7



The InGaP subcell has the best response to 500 nm wavelength light of the different types shown here.

We do not have the same goal as solar cell developers that want to capture a range of wavelengths.



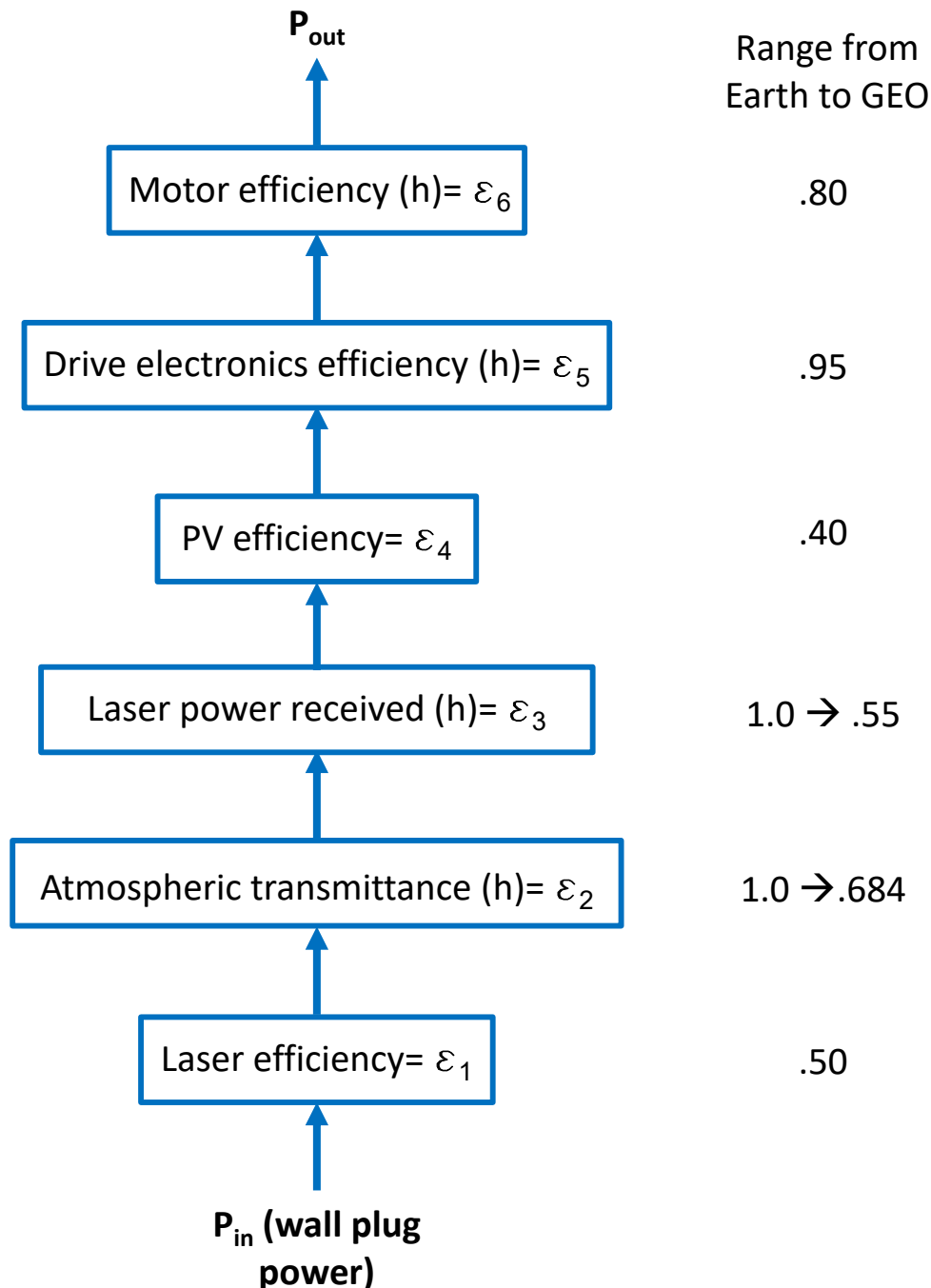
PV Array power characteristics

- Each half-climber carries its own PV Array
- Half of 11.2 MW of power is 5.6 MW of power per half-climber
 - Assumes startup can exceed the 4 MW limit
- The minimum diameter of the array on each half-climber is ~5 m, 19.63 m² area
- $5.6 \text{ MW} / 19.63 \text{ m}^2 = 290 \text{ kW/m}^2$
 - This looks like a feasible number for PV arrays
- The biggest issue is going to be keeping the temperature of the PV arrays in the 80°C-100°C range to maximize efficiency

The optimization problem

- We see that .5 micron wavelength light produces a beam that can transmit tens of thousands of kilometers without diverging too much
- .5 micron is a good wavelength for transmission through the atmosphere
- .5 micron may be a little short to get the best efficiency out of PV arrays
- A custom semiconductor mixture will be needed which may give as high as 40% efficiency

The transfer function to calculate the efficiency of the total system



$$P_{out} = \varepsilon_1 * \varepsilon_2 * \varepsilon_3 * \varepsilon_4 * \varepsilon_5 * \varepsilon_6 * P_{in}$$

The total system efficiency is a function of altitude as different effects happen at different altitudes.

When the climber is at the Earth, the system efficiency is ~15%.

If we need 11 MW of power on the climber at Earth, we need to put 73 MW of wall power into the system. Naval reactors can provide this.

If we limit the total power to 4 MW, then we need 27 MW of input power.

At GEO, the system efficiency is ~5.7%. The climber needs very little power the closer we get to GEO.

Another issue to consider with laser power beaming: IR reflectance from the PV arrays

The climber is shown about 30 m above the anchor platform deck. This reflection effect is highly dependent on the locations of the lasers and the altitude of the climber. At higher altitudes, the reflected beam may be much further away from the platform due to transverse oscillations of the tether, but will also be larger and more diffuse.

Reflected IR light from climber could create hot spots on deck. Safety measures would be in place to protect workers

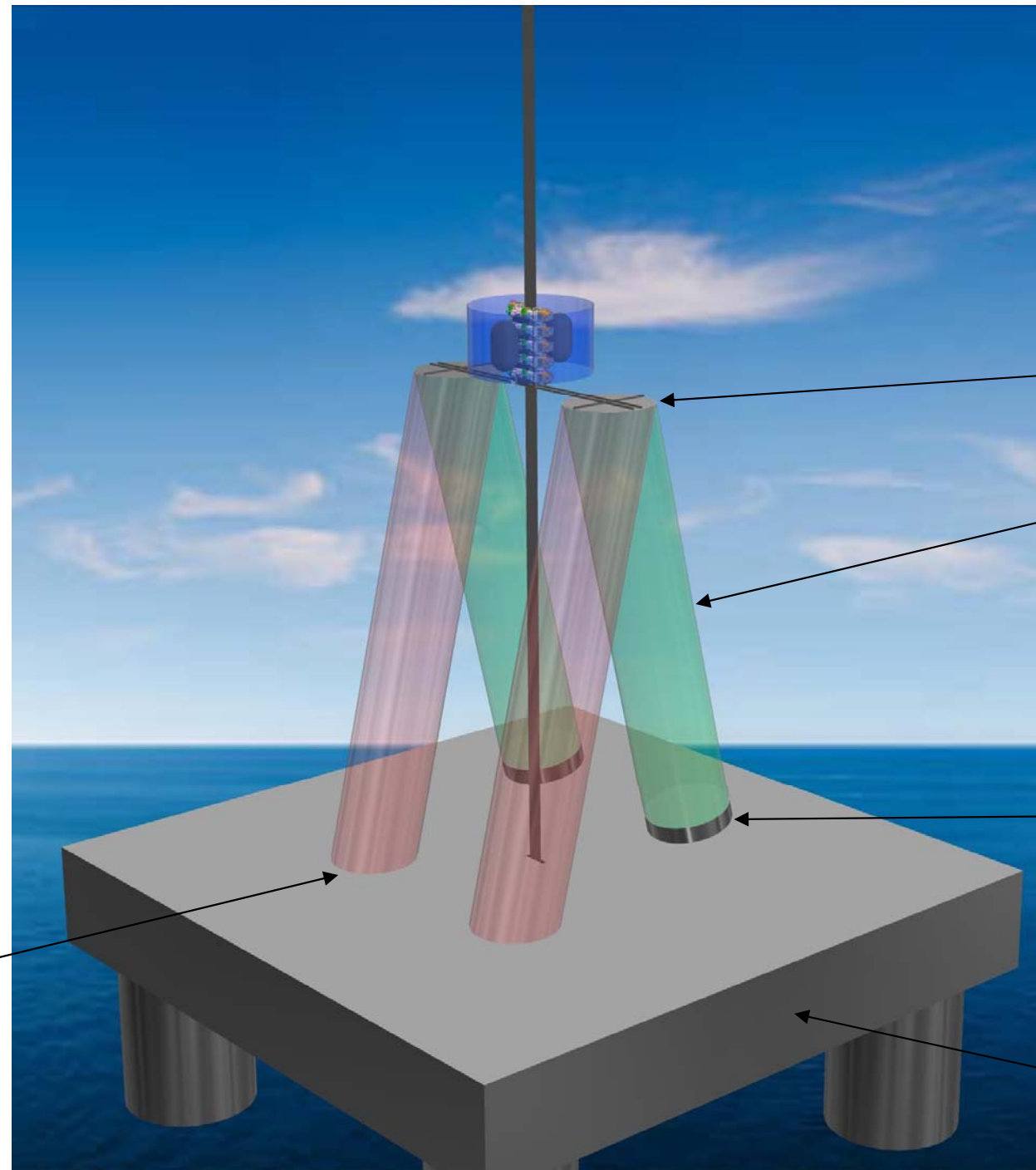
Green laser light can also be hazardous to eyes. Much care will have to be taken.

5m diameter PV array

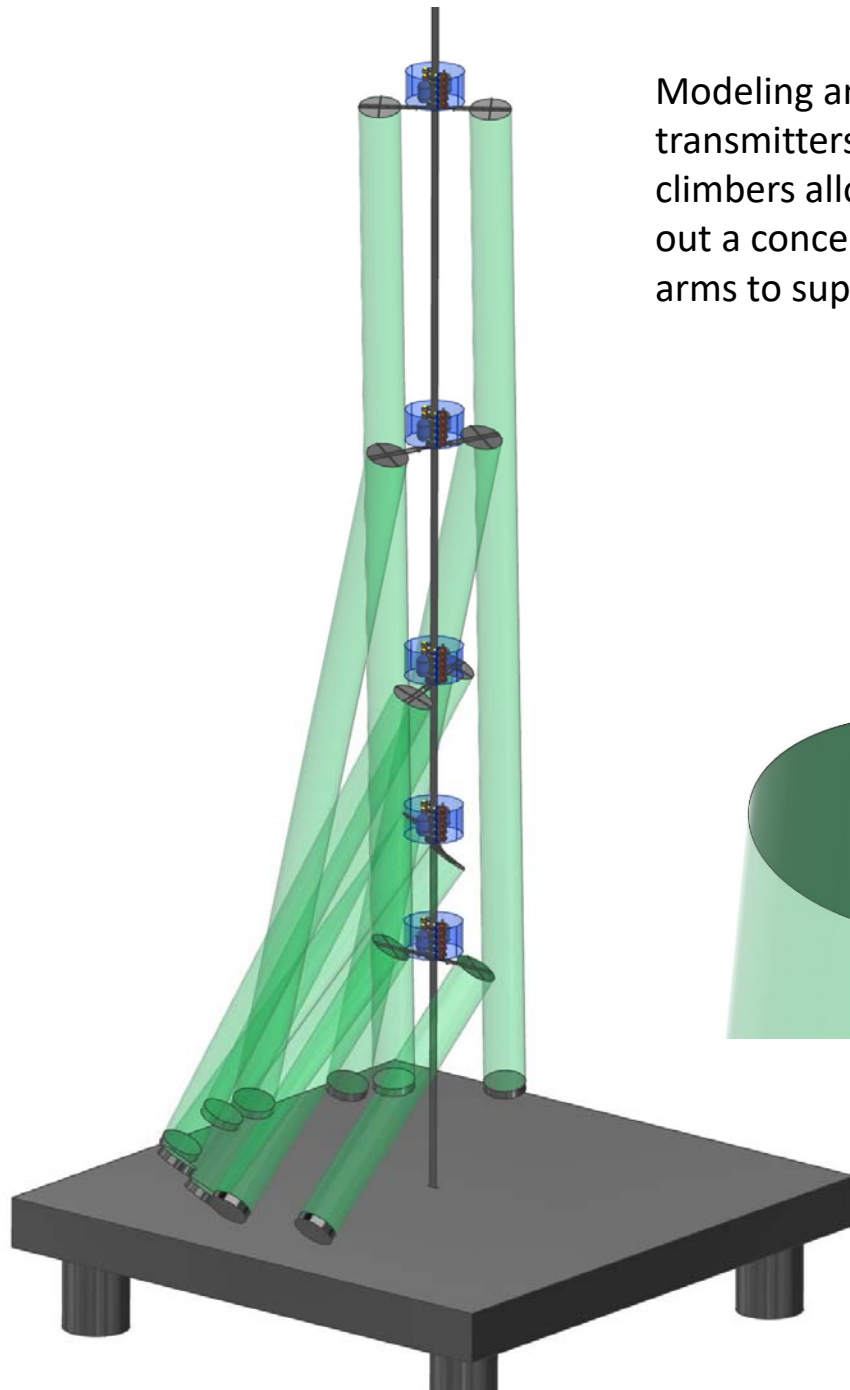
Laser light to climber

5m diameter laser transmitter

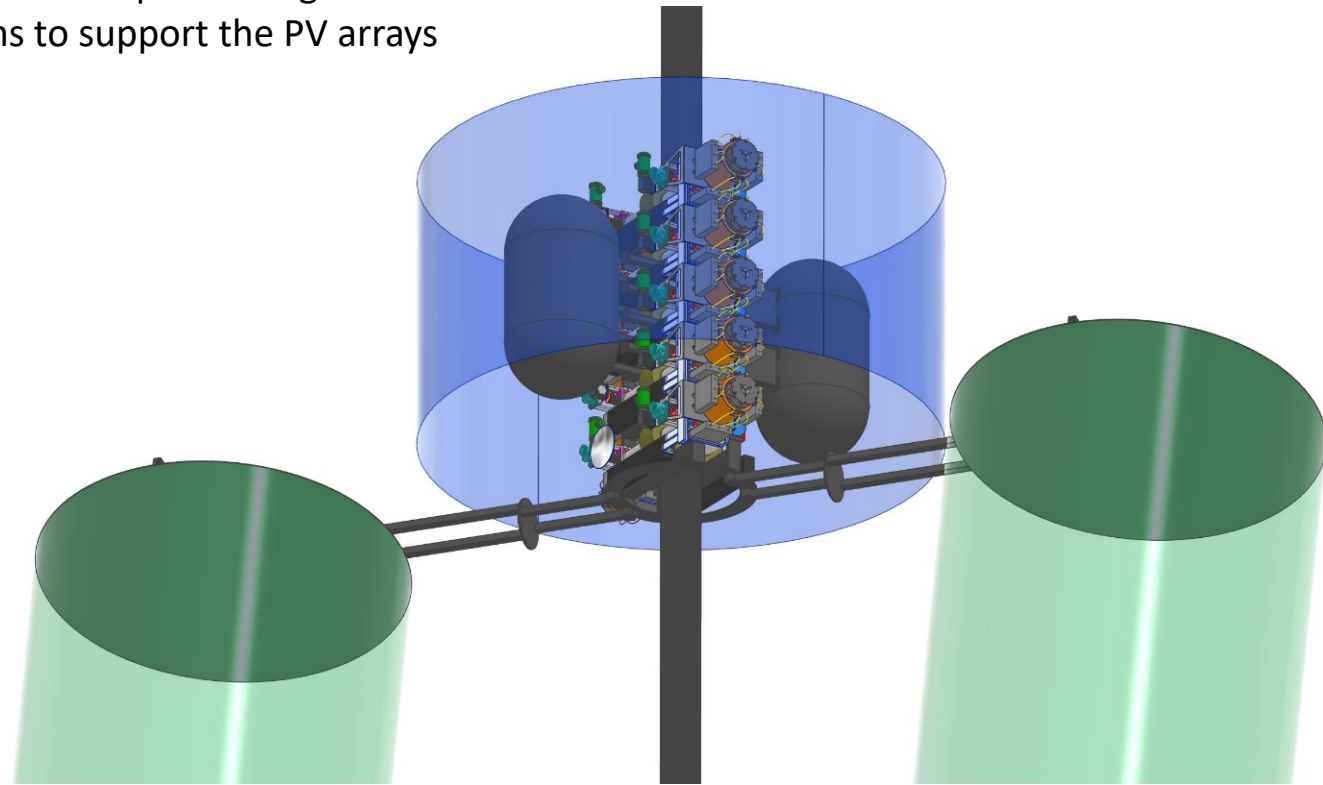
Earth anchor platform



Things get a bit more complicated when beaming power to multiple climbers



Modeling an array of laser transmitters to power multiple climbers allowed me to flesh out a conceptual design for the arms to support the PV arrays



I discovered recently that power beaming to multiple climbers has an issue that nobody has discussed before: How do you keep the PV arrays from shadowing each other?

Conclusions

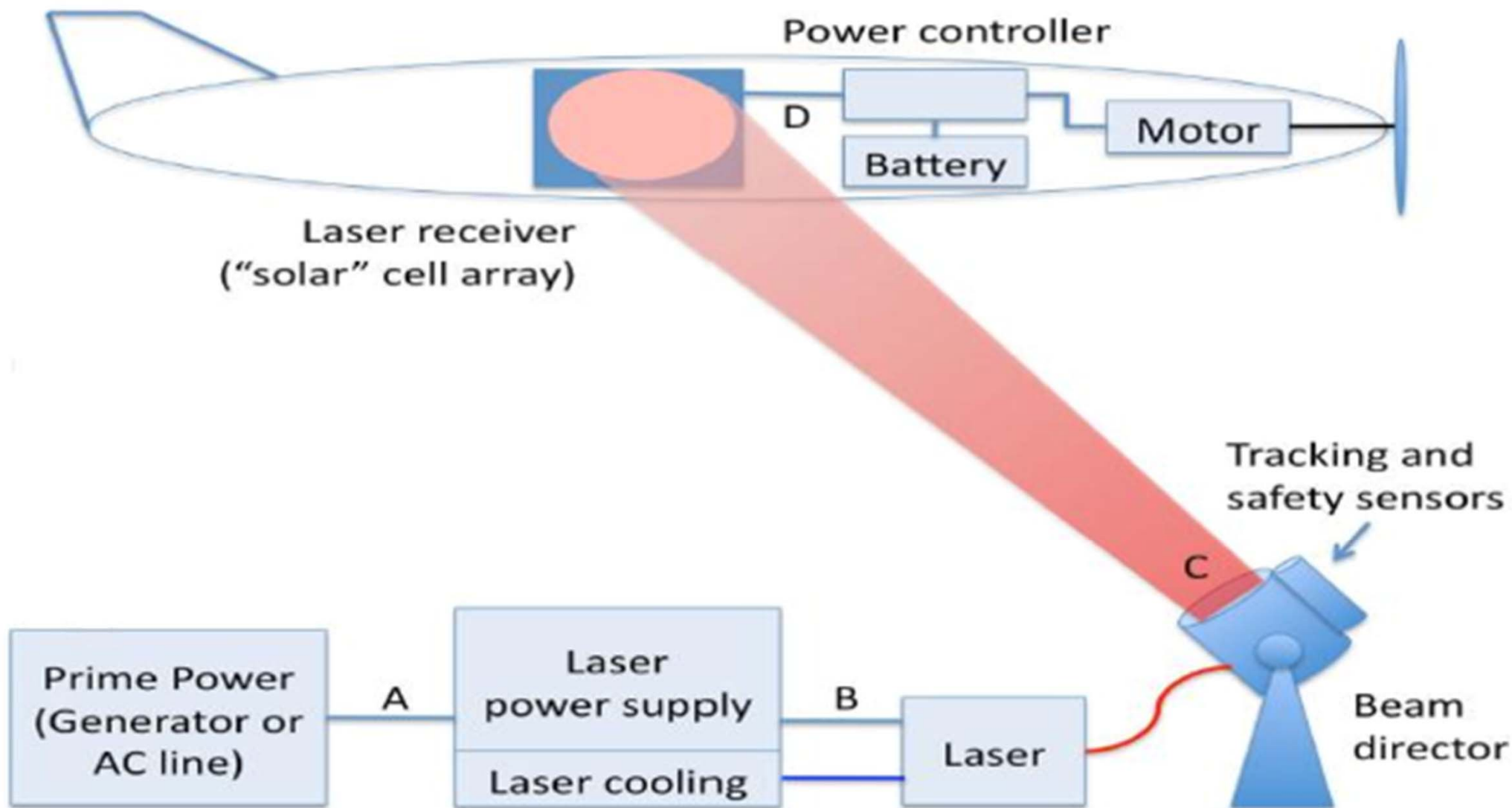
- Because 99.7% of the laser's trip is in vacuum, choosing a wavelength around $.5 \mu\text{m}$ and a beam waist of $\sim 5 \text{ m}$ diameter should transmit with sufficient efficiency
- Much more work is needed to settle on the best laser and PV technologies to combine into the power beaming system
- A nuclear reactor (like a submarine's) may be the best continuous provider of $>50 \text{ MW}$ of wall plug power to the lasers
- Trade-off between space-based and ground-based lasers needs to be studied, as well as solar conversion at high altitude

References

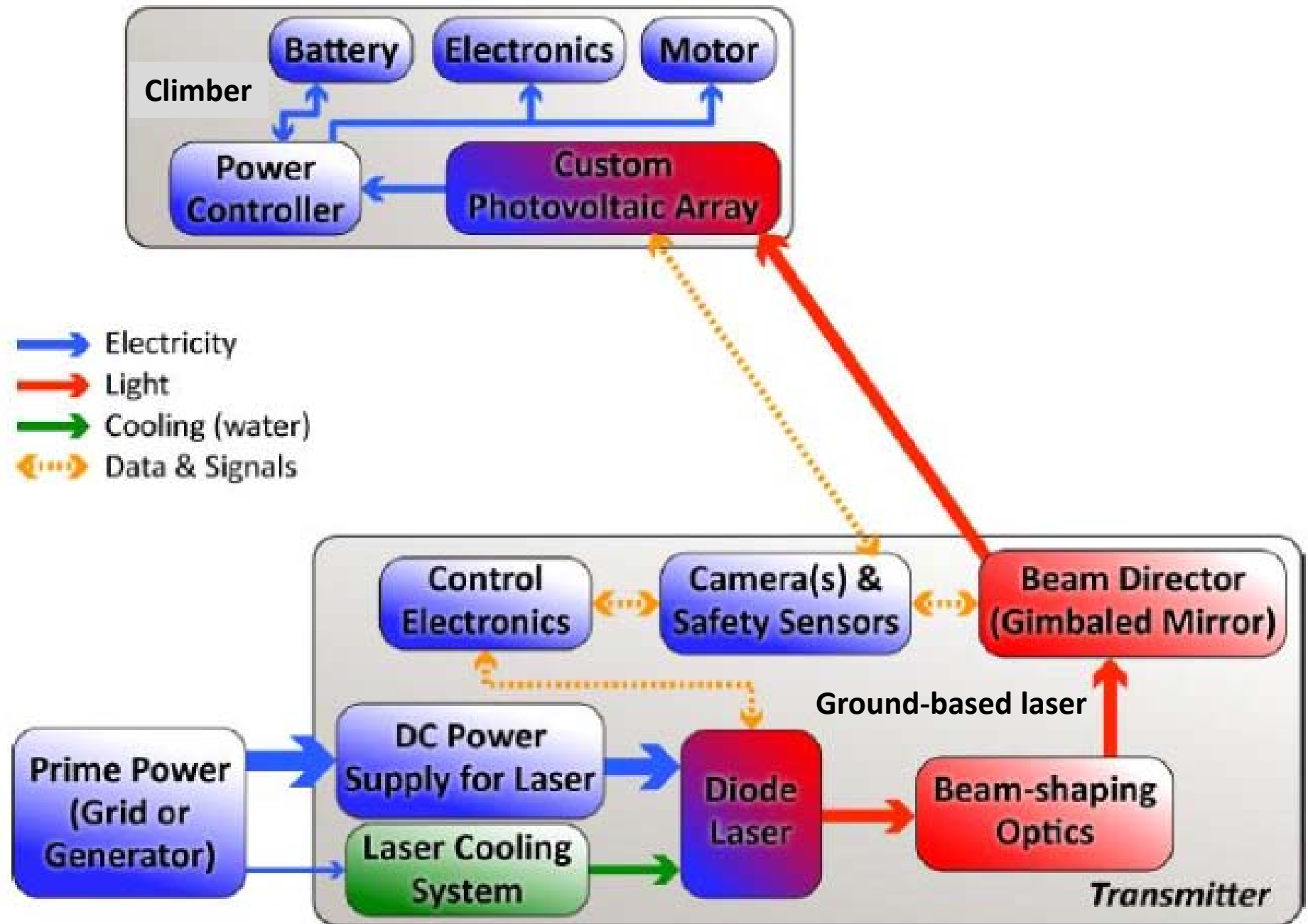
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BACKUP SLIDES

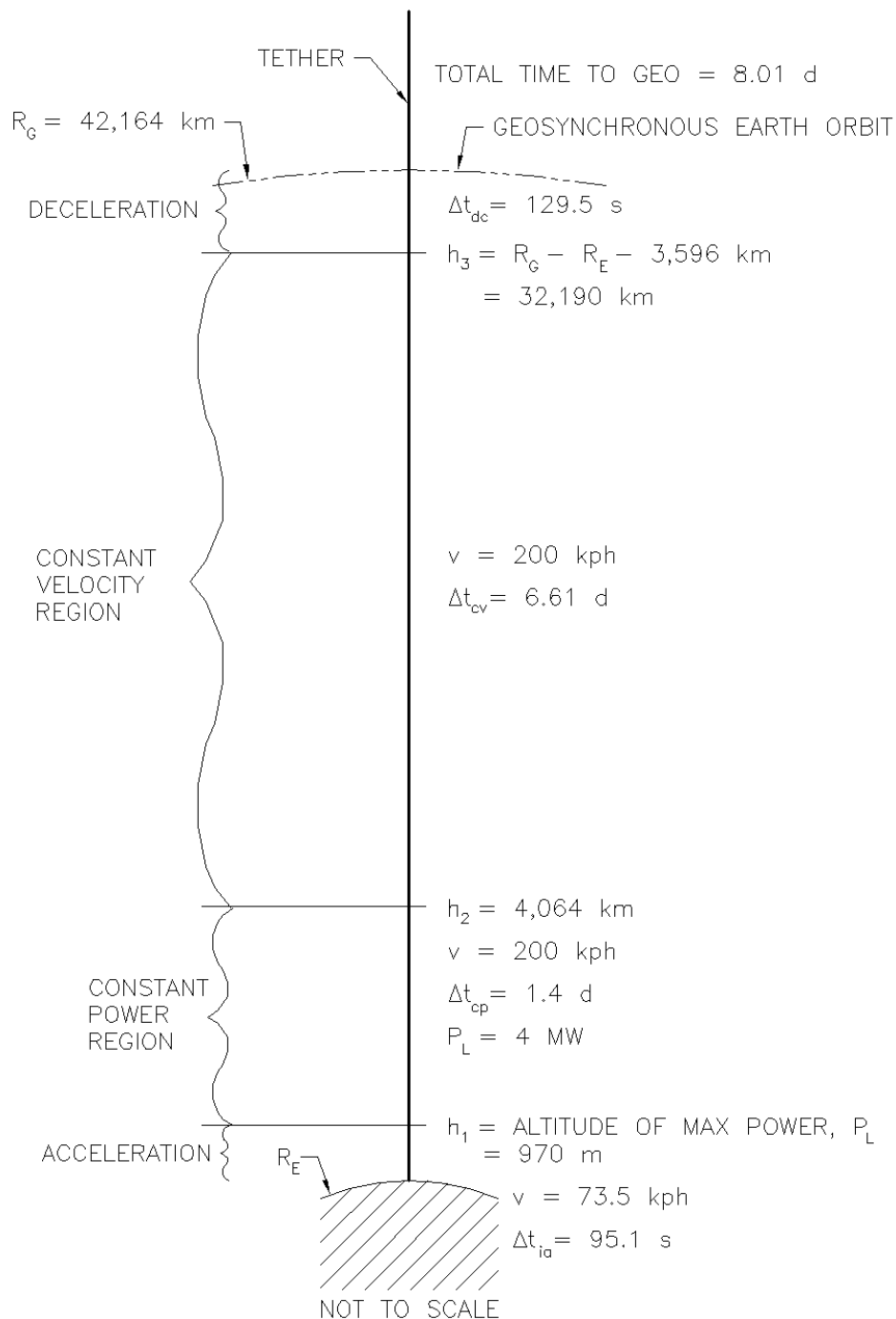
Wireless power delivery to an aircraft or UAV



Block diagram of a laser power beaming system



Power vs Altitude ranges factoring in acceleration, deceleration and 4MW limit



The equation for the mechanical power output of the climber is:

$$P = F \cdot v$$

- P is the power
- F is the weight of the climber
- v is its linear velocity.

This is not the same as the electrical energy that went into the motors.

When the climber is just starting out, $v=0$, so the above equation says that the mechanical power to lift the climber is zero.

The electrical power required to start the motors turning is at its maximum at this point, and is far more than the mechanical power calculated by the above equation.

From the news...(see backup slides)

...it's clear that we don't have multi-megawatt lasers in the size range we need, off the shelf.

- Military lasers do not have the characteristics we need because they are not intended for the range we need to transmit power. (Their targets are closer.)
- They also don't care how hot the surface of the target gets
- Assume 50 kW CW lasers (like welding lasers)
 - If we need 4-6 MW of laser power per transmitter then we need up to 120 lasers ganged together and the combined beam sent through a beam expander

Britain's Dragonfire laser weapon engages first aerial targets

By David Szondy
January 21, 2024



The British government isn't saying much about Dragonfire, though some things are known. It's a solid-state laser weapon in the 50-kW range based on doped glass fiber bundles running into a British-designed beam-combining system that's still very much on the classified list. Mounted in a turret, there's also an electro-optical camera and a secondary laser for target acquisition and beam focusing.



Northrop Grumman Laser Weapon System Completes Deployment on USS Portland

Posted on April 6, 2022 by Richard R. Burgess, Senior Editor



Amphibious transport dock USS Portland (LPD 27) transits the Gulf of Aden, Dec. 13, with a Solid State Laser – Technology Maturation Laser Weapons System Demonstrator Mark 2 MOD 0 on board. The Office of Naval Research selected Portland to host the laser weapon technology in 2018. *U.S. MARINE CORPS / Lance Cpl. Patrick Katz*

NATIONAL HARBOR, Md. — The Laser Weapon System Demonstrator deployed on the U.S. Navy's amphibious platform dock ship USS Portland (LPD 27) has completed its first deployment as the Portland returned to its homeport of San Diego in March.

Donna Howland, Northrop Grumman's acting business development director for Directed Energy and program manager Laser Weapon System – Demonstrator, told *Seapower* April 6 the Navy said it was "able to exercise the high-energy laser in the 5th Fleet in December

The **150-kilowatt** LWSD [Laser Weapon System Demonstrator] is mounted on the superstructure of the Portland and is integrated with the ship's combat information center, where a control console is installed. Northrop Grumman made the Tactical Laser Core Module of the system, while the U.S. government made the system's energy and thermal storage modules.

<https://seapowermagazine.org/northrop-grumman-laser-weapon-system-completes-deployment-on-uss-portland/>

From: <https://optics.org/news/14/11/9>



daily coverage of the optics & photonics industry and the markets that it serves

Business News

nLight agrees megawatt laser weapon deal

07 Nov 2023

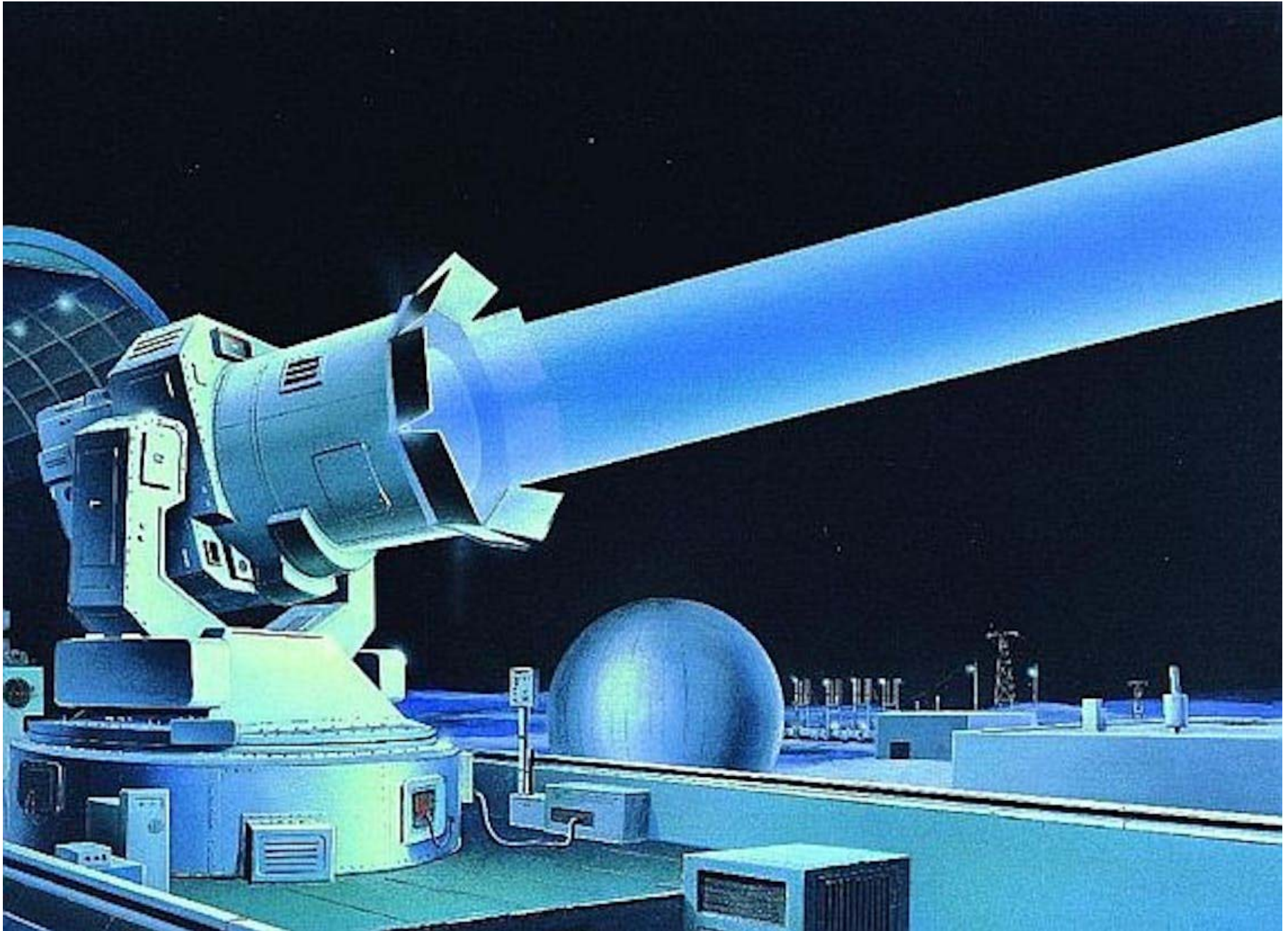
US Department of Defense award doubles value of existing contract to \$171 million.

The high-power diode and fiber laser manufacturer **nLight** says that it will now receive \$171 million from the US military to develop a **megawatt-scale laser weapon** - double the total originally awarded.

Back in May the Camas, Washington state, company announced **details of an \$86 million contract** with the Office of the Under Secretary of Defense (OUSD), for its High Energy Laser Scaling Initiative (HELSI) project.



Artist's rendering of a megawatt military laser



<https://www.militaryaerospace.com/sensors/article/16722085/military-eyes-prototype-megawattclass-laser-weapon-for-ballistic-missile-defense-in-next-seven-years>

MILITARY

Lockheed Martin juices up its laser weapon to half a megawatt

By David Szondy
July 31, 2023



<https://newatlas.com/military/lockheed-martin-boosting-laser-weapon-500-kw/>



Beamed-Energy Propulsion (BEP) Study

This study did a conceptual design on a ground-based laser system for propelling spacecraft to and in LEO. They are talking about >20X more laser energy than we need.

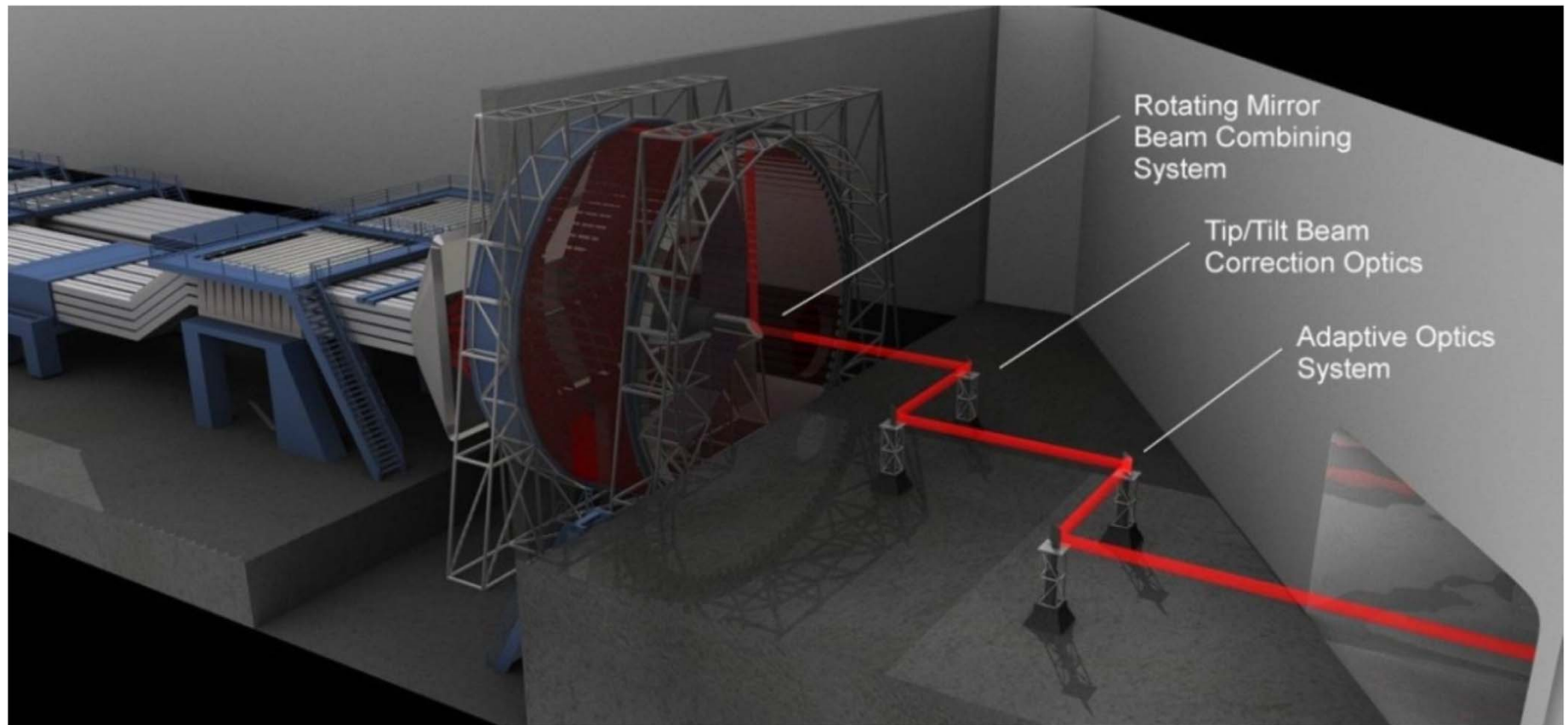
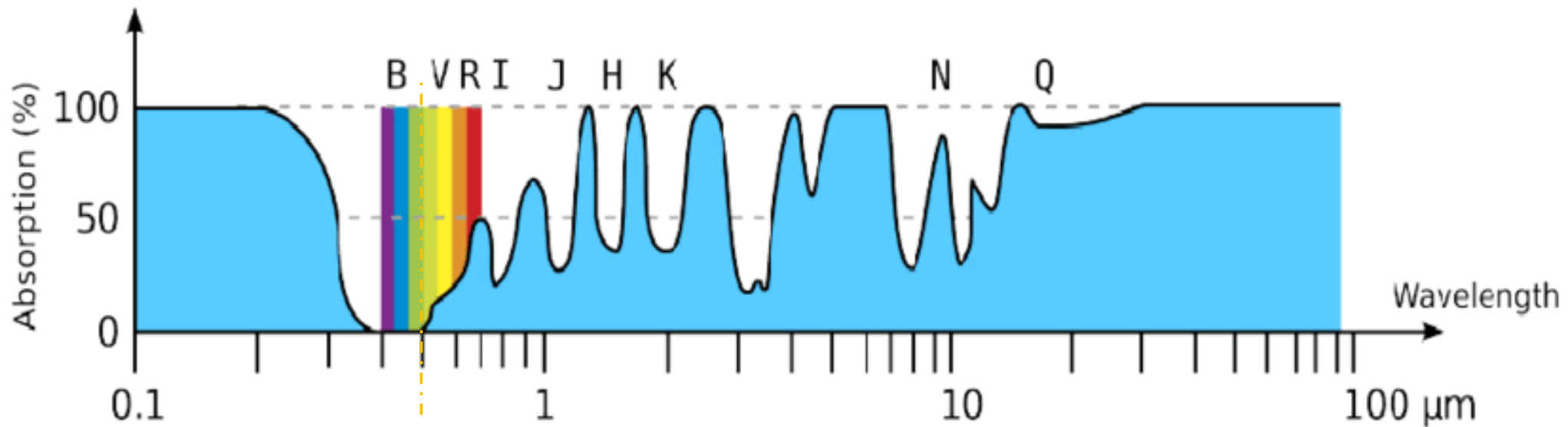


Figure 2.10.—Concept for beam-combining system with central rotating output mirror.

The boost laser consists of a continuous or a pulsed **140-MW** (Figure 2.9) output laser constructed of **up to 2900 (continuous or pulsed), 120-kW** commercial-off-the-shelf (COTS) modules. Two options exist for the “main” laser: (1) a ground system of up to **7250 COTS lasers (20 kW each)** or (2) a space-based relay station that utilizes the boost laser facility.

Absorption of light by the atmosphere as a function of the wavelength of the light (microns)



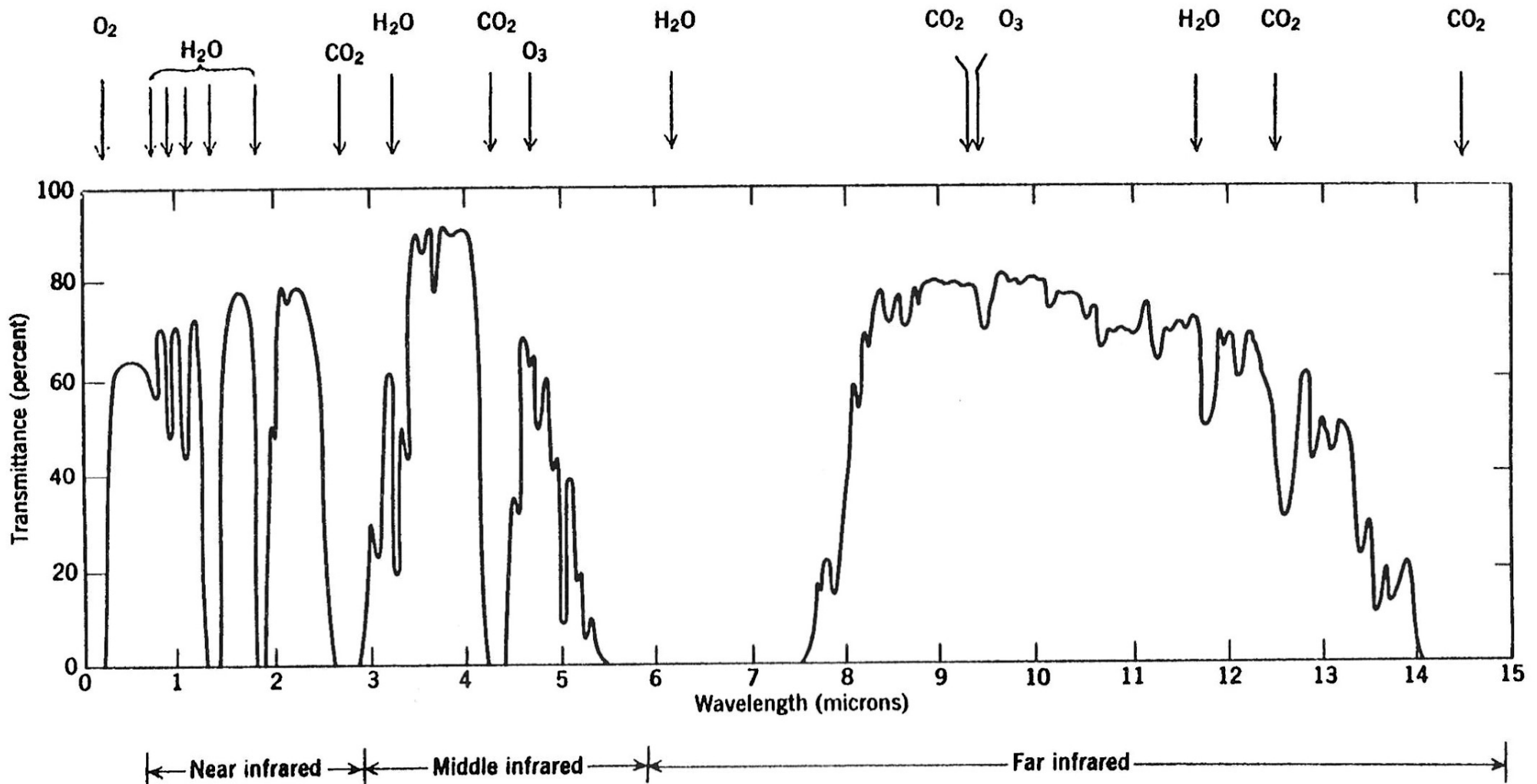
https://www.researchgate.net/figure/Earths-atmospheric-absorption-as-a-function-of-wavelength-Adapted-from-2_fig1_286165361

This graph is a major reason for choosing .5 microns for the wavelength of the laser shown in the next section. This is a green laser in a region of the chart where there is the least absorption from the atmosphere.

The wavelength of .84 μm is in the middle of the range of sensitivity for GaAs PV cells, but this IR frequency is more absorbed by the atmosphere than the shorter green wavelength.

Note this is a log scale in wavelength.

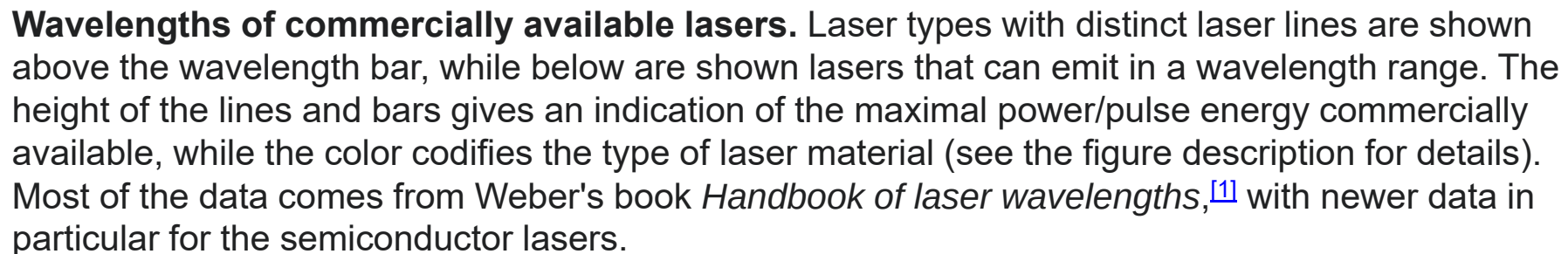
More detailed transmittance map of infrared by wavelength



Linear scale in wavelength, y axis is Transmittance (Previous graph was absorption.)

Fig. 2.1. Atmospheric transmittance measured over 1820-m horizontal path at sea level. (From R. D. Hudson, Jr., Infrared System Engineering, Wiley & Sons, 1969, p. 115.)

As shown later, the best wavelength for long range transmission is around .5 microns, green visible light



Compact FEL designed at JLab

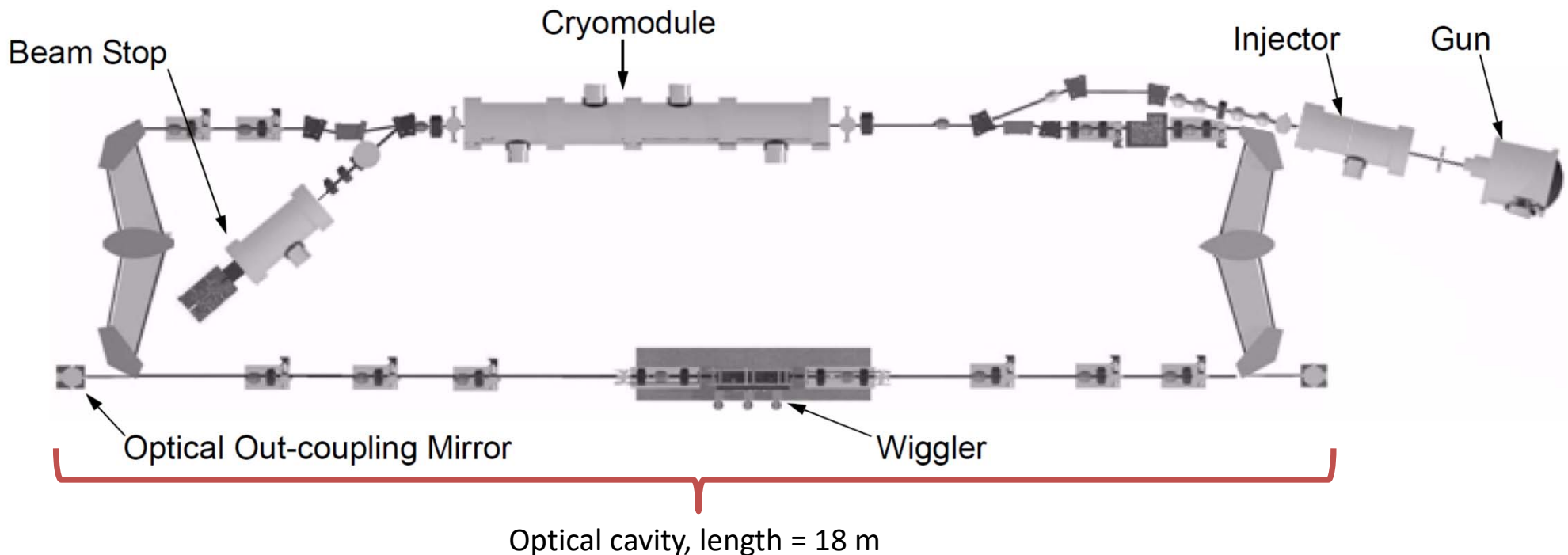
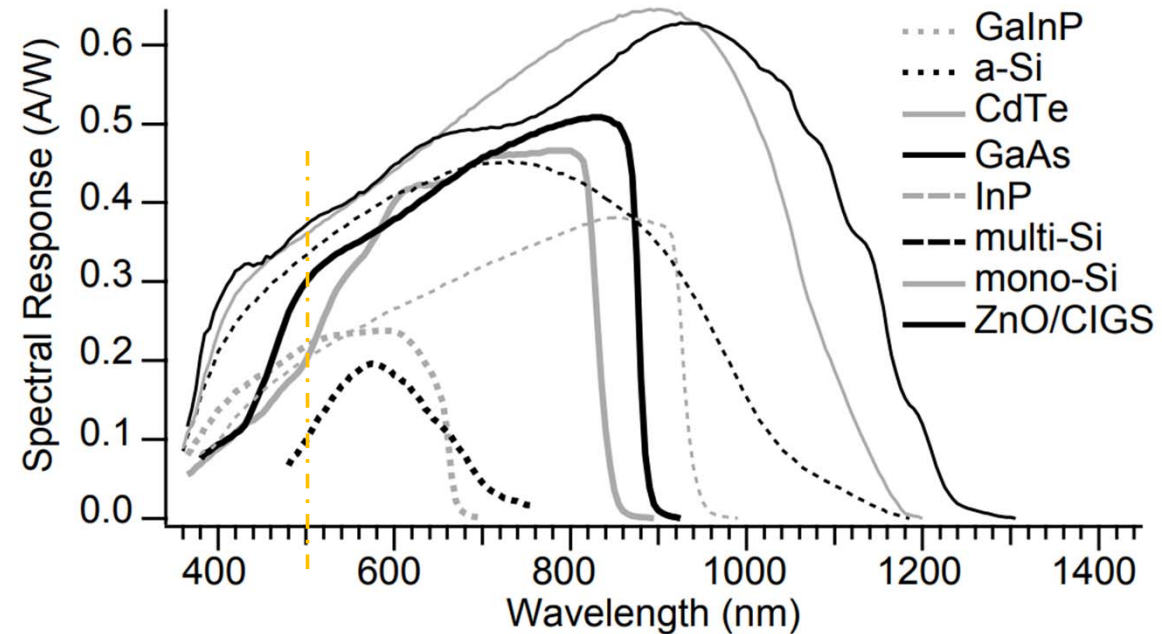


Figure 2: Advanced accelerator physics optics concepts were used to achieved the design of a compact FEL. The total length of the optical cavity is 18 meters. The system design is for MW and higher capabilities in the extracted optical beam. The injector, cryomodule and beam stop all have SRF cavities. The electron beam from the photocathode is accelerated to 0.5 MeV in the DC gun and then to 7.5 MeV in injector. The cryomodule accelerates the beam to 80-160 MeV depending upon the wavelength of light desired. The high-power optical beam is created in the wiggler. Approximately 99% of the electron beam's energy remains after the wiggler and using ERL principles, it is recovered by passing the beam with reverse phase back through the cryomodule. An SRF cavity in the beam stop removes most of the energy that was supplied by the injector. This RF power is returned to the Injector. A small amount of energy is absorbed at the end of the beam stop. (Ref. 17)

PV Receiver Technologies, 3

- **Three types of Thin film PV modules:**
 1. Amorphous Silicon (a-Si) thin-film
 2. Cadmium Telluride (CdTe) thin-film
 3. Copper Indium Gallium Selenide (CIGS)
- Thin-Film solar panels are less efficient and have lower power capacities than mono and polycrystalline solar cell types. (Ref. 8)
- The efficiency of the Thin-Film system varies depending on the type of PV material used in the cells but in general they tend to have efficiencies around 7% and up to 18%. (Ref. 8)
- Thin-Film solar panels have a better temperature coefficient than silicon based panels. They are less affected by high temperatures and will lose only a small portion of their performance when it gets too hot. (Ref. 8)
- 19% efficiency conversion, (Ref. 4)
- Not a good fit for the climber

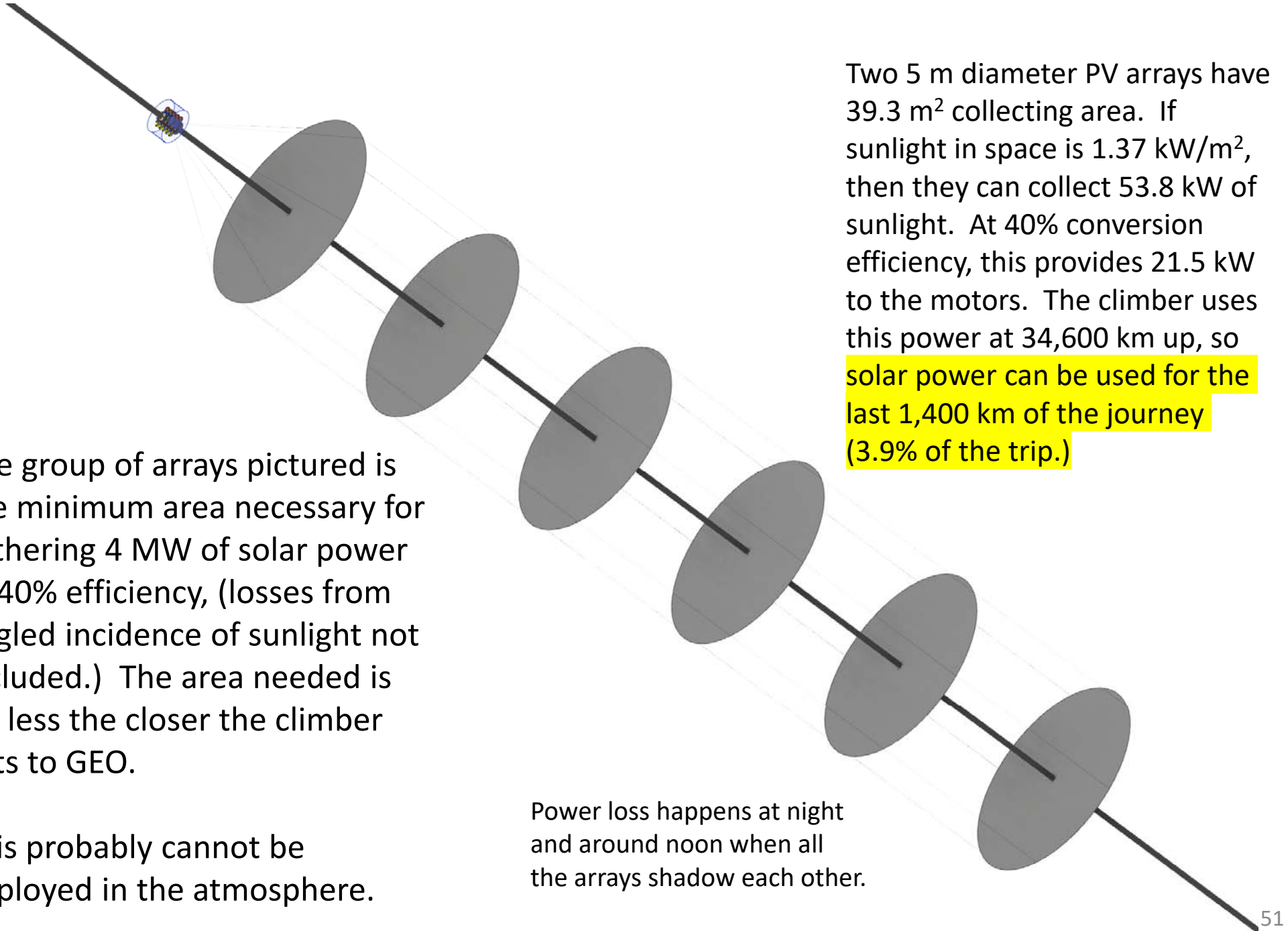


Spectral response peaks well above 600 nm for all three types of thin film PV cells

Comparison with solar arrays

- A 20 MW solar farm on Earth needs ~68 acres (.275 km²) of land area
- In space, solar power available is 1.37 kW/m²
 - To get 4 MW of power requires 2,920 m² of PV area at **100%** efficiency
 - At 40% efficiency, this becomes 7,300 m² of PV area
 - Dividing this area into two circular PV arrays, they are each 68.2 m in diameter (compared with 5 m diameter PV arrays for laser systems)
 - If I make 6 layers of arrays for the whole 7,300 m², each one is 39.4 m in diameter (see next slide)
 - Based on 2013 ISEC climber study report

Concept drawing of 20 tonne climber with six 39.4 m diameter PV arrays



Two 5 m diameter PV arrays have 39.3 m^2 collecting area. If sunlight in space is 1.37 kW/m^2 , then they can collect 53.8 kW of sunlight. At 40% conversion efficiency, this provides 21.5 kW to the motors. The climber uses this power at $34,600 \text{ km}$ up, so solar power can be used for the last $1,400 \text{ km}$ of the journey (3.9% of the trip.)

The group of arrays pictured is the minimum area necessary for gathering 4 MW of solar power at 40% efficiency, (losses from angled incidence of sunlight not included.) The area needed is far less the closer the climber gets to GEO.

This probably cannot be deployed in the atmosphere.

Power loss happens at night and around noon when all the arrays shadow each other.

Reasons to put the lasers for the SE at ground level:

1. The highest power is required at the ground. Put a concentrated power source where it can be cooled easily with convection and conduction (like a nuclear reactor.)
2. Diffraction naturally reduces power with altitude (works against power being beamed down).
3. Need smallest PV arrays because the arms for the array must support the full weight of the array since they are not in zero G. Stronger arms mean less payload.

Common “alternatives” to ground-based laser for the space elevator:

1. Beaming power down from GEO with solar-powered lasers
 - a. Enormous PV arrays in GEO because of low power density of solar radiation in space
 - b. Long distance down to Earth means diffraction-expanded beam at Earth bathing the Earthport in hazardous green light
2. Beaming power down from LEO with a swarm of solar powered laser satellites
 - a. Satellites only have a few minutes to aim their lasers at the climber
 - b. Need a lot of satellites (which compounds the congestion of LEO and raises the probability of collision with the tether).
 - c. Gigantic solar arrays in LEO
 - d. Hazardous green light bathing the Earthport
3. Electrical powerline up to some altitude which then falls back to Earth
 - a. Need powerline carrying kiloamps of current that is flexible and light. Probably can't make one for more than a few kilometers at best.
 - b. How to lower it safely for reusability?
4. Using the tether as a conductor for electricity from GEO solar power satellites
 - a. See my calculation on how lossy this is
5. Battery powered through the atmosphere, then solar powered above
 - a. State of the art lithium sulfide batteries on the climber now can give full power for 5 minutes
 - b. Arrays too large because sunlight is too diffuse

Something to think about...

- From: <https://physicsworld.com/a/megawatt-laser-beacon-could-communicate-with-aliens/>

“Clark and Cahoy discovered that a 2 MW signal fired through a 30 m telescope would have a large enough magnitude to clearly stand out from the Sun’s natural variation in infrared emissions to astronomers on Proxima Centauri b. At the same time, a 1 MW laser combined with a 45 m telescope would not only be clearly detectable in TRAPPIST-1, but would be visible at 20,000 light-years away – covering the entire Orion Arm of the Milky Way.”